

Big data ethics: implications for organizational performance and business value

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Abstract

Purpose – Ethics has a significant impact on the decisions of organizations at all levels. The ethical values related to big data in business organizations have received limited research attention, except for a few notable examples. According to the literature, once an initiative is taken to create and disseminate a code of ethics, emerging obstacles begin to set in. Hence, the primary purpose of this study is to map, visualize and analyze the development of research in the area of big data ethics, focusing on the industry in general, with specific emphasis on organizations' performance and business value.

Design/methodology/approach – To achieve the goal of the study, we explored the Scopus database, focusing on research related to big data ethics in business. We retrieved 277 studies based on predetermined keywords approved by the authors. The time frame of the published studies spans from 1990 to the present. Biblioshiny and VOSviewer were used to analyze the retrieved studies (273 studies). Following that, the studies were examined using a strengths, weaknesses, opportunities and threats (SWOT) analysis, focusing on the main topic of the study and referring to the surveyed literature. Finally, to discuss the results and research implications, we utilized the theory-context-characteristics-methods (TCCM) framework.

Findings – The theme of “ethics” contributed to the development of several research areas, including privacy, data governance, information dissemination, human-based research and behavioral research. The study highlights a gap in the existing literature regarding the adoption of metaethical and normative ethical lenses in big data research.



Originality/value – The study discusses diverse industry examples, such as tourism and healthcare, to illustrate the practical implications of ethical practices while addressing the complex interplay between data governance, privacy and human rights issues.

Keywords Ethics, Bibliometrics study, Big data analytics, Industry 4.0, SWOT analysis, Decision-making, Industrial innovation

Paper type Research article

1. Introduction

The rise of digital technologies has led to ethical concerns that have attracted the attention of both the general media and a growing number of scholars from various fields (Di, 2023; Munn, 2022; Rodgers *et al.*, 2023). In recent years, there have been several well-publicized ethical issues affecting the industry, businesses, and organizations. The general public expects businesses to adhere to rigorous ethical standards before being granted licenses to operate. During the 1980 and 1990s, surveys conducted by the Conference Board in the United States, the Institute of Business Ethics in the United Kingdom, and Klynveld Peat Marwick Goerdeler in Canada reported a rise in the number of companies with business ethics codes. According to these surveys, more than half of organizations now have formal codes of ethics and ethics programs (Webley, 2001). In the twenty-first century, the value of data has grown dramatically, often compared to the value of oil in the twentieth century (Krotov and Johnson, 2022).

The value of data has been amplified by the rapid and substantial growth of data-based tools and innovations that deploy various techniques, including Artificial Intelligence, Machine Learning, Deep Learning, and Data Mining (Olszak and Zurada, 2019). Driven by the availability of social media and online platforms, huge volumes of data have been generated, posted, reposted, reported, crawled, and analyzed. Hence, the quality of the data, along with the related management strategies at both the organizational and individual levels, has received increasing attention (Alshawi *et al.*, 2011; Liu *et al.*, 2023; Santoro *et al.*, 2018). However, understanding the quality of data requires a clear understanding of the different dimensions and the criteria used to evaluate each one (Caballero *et al.*, 2022). Examining the dimensions that frame the quality of data will allow businesses to implement the necessary strategies that will, in turn, enhance their performance (Fan, 2015). Following suitable appraisal strategies for evaluating data quality can ensure that the data meets the aims and vision of the organization. To ensure that the data meets the required level of quality, it should be repeatable, defensible, and usable. Additionally, the organization should be able to identify errors in the data and address them with appropriate policies (Recknagel and Michener, 2017). Based on this, quality control and quality assurance themes have emerged to ensure that data meet the required level of quality in different application domains (McCord *et al.*, 2022).

The academic community's concern about the ethical deployment of AI research to support advanced technologies has been highlighted in the literature. Hence, researchers have presented various perceptions regarding related topics, including biased models, deepfakes, addictive behaviors, misinformation, and the discriminatory deployment of emotion detection and facial recognition models. Ethics in AI is more important than ever (Dattner *et al.*, 2019; Holmes *et al.*, 2021), knowing how much society is impacted by AI applications and tools. More and more data scientists contend that moral principles can and should guide AI research and applications, particularly in the context of social and political objectives. The importance of academic freedom, technological advancement, and value neutrality is emphasized by another group of data scientists. Particularly in the data science field, there is a significant ideological divide between atomist and holist ideologies (Greene *et al.*, 2023).

Ethics is a crucial consideration in the use of big data (Metcalf and Crawford, 2016; Xafis *et al.*, 2019), as it involves the collection, analysis, and use of large amounts of personal and sensitive information. Big data often involves the collection and analysis of personal information, making it essential to ensure that this data is collected and used in ways that respect individuals' privacy rights (Abouelmehdi *et al.*, 2018). As the use of big data becomes increasingly pervasive in modern society, ethical considerations are becoming more

important. Ethical considerations can help build trust between data collectors and data subjects, as well as between data users and society as a whole. This trust is essential for the continued growth and development of the big data industry. Big data ethics is a set of moral principles that govern the collection, analysis, and use of large amounts of data.

Organizations have utilized big data in different sectors to improve their performance and gain a competitive advantage within the market (Ghasemaghaei, 2020). Still, the deployment of big data has been linked to several ethical concerns. The deployment of big data within organizations' procedures should balance the expected benefits with ethical considerations (Tanggamani and Alias, n.d.). This can be facilitated through skilled employees, sophisticated tools, and proper ethical policies. Despite the increasing attention from researchers, there is an evident research gap, highlighted by the need for a roadmap to address several ethical issues of big data utilization, including consent, anonymity, and privacy (Zimmer, 2018). This also calls for standardized ethical protocols to be followed within organizations while focusing on the impact of emerging technologies (Roche and Jamal, 2021).

Ethics in the big data era span across several dimensions of the technology itself, society, organizations, and countries. On the technological level, literature has called for an "ethics by design" approach, along with advancements in technologies (Mohammed *et al.*, 2021). On the societal level, several ethical concerns have been raised in the literature, calling for the inclusion of different stakeholders and responsiveness to societal needs (Eastwood *et al.*, 2019). On the organizational level, there is a call for designing and following ethical schemes, building on the "Ethical Theory of Organizing" proposed by Yazdani and Murad (2015). While data-enabled technologies in organizations offer several benefits, they also impose reputational and legal concerns regarding data privacy (Mohammed *et al.*, 2021). Regulating and managing access to massive data is a significant concern for businesses worldwide. On the country level, nations face major challenges when it comes to passing laws pertaining to data security and accessibility (Bag *et al.*, 2023).

Building on the above discussion, this research aims to explore the current body of literature retrieved from the Scopus database and analyze it using two main methods: bibliometric and SWOT (Strengths, Weaknesses, Opportunities, and Threats) analyses. The bibliometric analysis will help the researchers visualize the development and themes of the research and highlight the more significant areas that have been explored in the literature. On the other hand, SWOT analysis will allow the researchers to appraise the current body of knowledge, focusing on both the positive and negative views, as well as the internal and external factors that impact the management of ethics in big data. Furthermore, the Theory-Context-Characteristics-Methodology (TCCM) framework will be used to structure the discussion and research implications of the current study (Paul and Feliciano-Cestero, 2021).

The rest of this study is structured as follows: The Literature Review is presented in Section 2, the research method in Section 3, data extraction in Section 4, the bibliometric analysis of the surveyed studies in Section 5, the SWOT analysis in Section 6, and the research contributions and conclusion in Sections 7 and 8.

2. Literature review

2.1 Theoretical grounds

When people use the internet, whether actively (via blogs or social media posts) or passively (through cookies or search engine queries), they leave behind rich and valuable psychological and social digital footprints (Speckmann, 2021). At the individual, organizational, national, and even global levels, these digital footprints can enhance the understanding of various processes, behaviors, relationships, and interactions in the real world (Krotov and Johnson, 2023). Utilizing this knowledge at the organizational level can lead to an improved understanding of consumers, enhance existing business processes, support decision-making, guide organizational strategies, and generate rich information about products; ultimately enhancing the quality of goods and services (Grover *et al.*, 2018). A research question related to an

organization's internal or external environment serves as the foundation for the data science process, laying the groundwork to improve organizational performance and provide value to consumers. However, efforts must ensure the ethics and legality of data science processes within the organization as a whole. Integrating ethical values into digital transformation generally involves several factors, including responsible data use, social fairness, operational openness, and the avoidance of biased algorithmic conclusions (Brodny and Tutak, 2025).

Within ethics, there are generally three distinct ways to approach an ethical dilemma: deontological methods stressing laws and obligations, teleological ones focusing on the need to assess consequences, and others related to virtues (Ananny, 2016). Ethics in big data have been explored based on different theoretical grounds. According to Morán-Reyes (2022), moral philosophy can be classified into three main themes: (1) metaethics, (2) applied ethics, and (3) normative ethics. Metaethics falls within the concepts of ethics, their descriptions and meanings, and their grounds or causes. It primarily focuses on addressing questions related to the existence of knowledge regarding ethics and ethical truth. It also raises vital queries about the relevance, universal applicability, and subjectivity of ethics (Van Roojen, 2015). Applied ethics focuses on specific practical issues that should be addressed within daily life in a particular field. Normative ethics focuses on the judgments of what is considered correct or wrong behavior, which stems from our responsibilities toward others. All these themes should be carefully considered when studying ethics in big data, particularly regarding organizations' policies.

In the explored literature, big data ethics have been examined through these philosophical lenses. The majority of the literature has explored ethics in big data, building on the grounds of applied ethics. These studies have tried to provide practical solutions to address the risks and concerns linked with big data adoption. For instance, Ryan *et al.* (2020) explored ethical concerns related to smart technology adoption across several dimensions, building on big data analytics and providing several practical implications. Shafiq *et al.* (2020) explored supply chain analytics, focusing on the adoption of social responsibility practices. The study highlights practical implications for ethics implementation in business, shedding light on the important role of big data analytics in this context. Additionally, Mohammed *et al.* (2021) suggested an optimized privacy-preserving data mining solution, aiming to protect individuals' privacy and overcome the issues with high-dimensional data. Focusing on agricultural big data analytics, Ryan (2020) explored ethical concerns related to big data adoption. The study offers practical implications related to informed consent and power dynamics. Forgó *et al.* (2021) focused on the privacy of social big data and proposed a framework integrating methods, organizations, and infrastructure for the effective implementation of an open platform for big social data. Focusing on creating a comprehensive plan for the responsible use of big data, Cuquet and Fensel (2018) explored both possibilities and hazards, especially in relation to social benefits, privacy violations, and discrimination.

While the majority of the studies have focused on providing practical solutions to address problems related to ethics in the big data era, they also build on metaethics grounds, as they try to provide definitions and descriptions of ethics in big data (Kuc-Czarnecka and Olczyk, 2020). Normative ethics has also been the foundation of several studies in the literature. Stefanizzi (2022) discussed the ethical implications of data centrism, particularly concerning privacy rights, transparency, and fair treatment in the use of predictive techniques. Morán-Reyes (2022) explored big data, focusing on normative and metaethics as the basic grounds for the implementation of big data in professional codes. However, there is an evident gap in the literature, as most previous studies have focused on the applied ethics theme when studying big data ethics, specifically within the organizational context.

2.2 Big data ethics in business

The adoption of data analytics within organizations has become a cornerstone of their development and growth in the competitive market, but it has also been linked to ethical concerns. There are several challenges to the applicability of big data ethics within organizations and businesses. These challenges range from the privacy of data being collected and utilized in

organizational processes to increased bias and inequality. These challenges are also linked to several obstacles that might hinder organizations from addressing them appropriately. One obstacle to the applicability of big data ethics within organizations is the source of data utilized to reach decisions. The entity that conducts the process of big data analytics usually obtains data from multiple sources, tries to find interactions and patterns in this data, and eventually reaches insights that can help in the decision-making process. The collection of the data, however, might be performed by other entities, making it difficult to adhere to ethical data collection standards. Another obstacle is that the process of big data analytics might be exploratory, in which the main goal of the analysis is not so clear to the data analysts, making it difficult to specify the main goal of the process clearly for those from whom the data is collected, and consent should be obtained. When securing participants' consent regarding data collection, it is important to clearly identify the possible risks and harms. However, it might not be clear what harm or risk could be triggered by advanced big data analytics to individuals or the overall community.

The ethics of big data in organizations have been explored across diverse contexts. In the mental health sector, [Rubeis \(2022\)](#) examined the ethics of big data and AI, focusing on three main dimensions: data mining, ecological momentary assessment, and self-monitoring. Data mining involves modeling and analyzing large volumes of data, while ecological momentary assessment refers to evaluating individuals' behavioral patterns within their natural environments. Self-monitoring, on the other hand, involves the active collection of data by individuals themselves. In the insurance sector, [Mullins et al. \(2021\)](#) investigated the ethical implications of big data across different stages of the value chain: product design, pricing, sales, and e-services. The study identified several ethical concerns, including the market exclusion of vulnerable customers, a lack of transparency regarding the benefits of data collection, and algorithmic bias. Within the agricultural domain, [Mark \(2019\)](#) highlighted key ethical challenges associated with big data, such as data accuracy, privacy, security, intellectual property, and risks to animal welfare and the environment. In the context of smart cities, [Mark and Anya \(2019\)](#) explored ethical issues related to smart information systems, emphasizing inequality, economic pressures, and bias. The study also addressed critical ethical factors such as accuracy, trust, transparency, and privacy.

2.3 Big data ethics and moral values

Big data is characterized by its volume, variety, velocity, veracity, and value ([Wamba et al., 2017](#)). The immense volume of big data, along with its complex nature, is linked to the collection, storage, and sharing of data from diverse sources such as technologies, mobile phones, and cloud computing. Big Data Analytics (BDA) approaches have been used to describe the collection, analysis, and processing of big data ([Wamba et al., 2017](#)). Despite significant advancements in BDA approaches, human engagement remains necessary in an interactive and iterative process. With this engagement, there is a call for moral and ethical principles to be followed at both individual and organizational levels. Data analysts, by virtue of their roles, have control over datasets and are expected to provide insights for decision-makers based on the data. Misinterpretation of analysis results can impact business performance ([Bag et al., 2023](#)). Furthermore, if data analysts lack moral discipline, the privacy of confidential data may be compromised ([Bag et al., 2023](#)). Accordingly, in this context, ethical behavior reflects the positive and conscientious perspectives and actions that data analysts follow toward the organizational culture, environment, and guiding principles, which are vital for meeting the organization's goals and enhancing its performance ([Grodzinsky, 2017](#)).

The ethics of big data have been explored, focusing on diverse moral values. Big data usage has been linked to privacy challenges in several studies. Big data are often distributed and connected across various data sources without the appropriate acquisition of users' consent regarding data usage, which poses serious risks to users' autonomy ([Salerno et al., 2017](#)). Reuse of personal data for purposes other than the original ones is one of the main risks to the privacy of information ([Herschel and Miori, 2017](#)). Another challenge arises from unauthorized access to users' private information ([Herschel and Miori, 2017](#)). Big data

have also been linked to social exclusion, neglecting the needs and requirements of minorities who are underrepresented in the data (Lerman, 2013). This is particularly an issue when analysis and decision-making processes within organizations depend on biased data. It also adds a new layer of inequality to existing narratives, further excluding minorities, preventing them from the efficient use of technologies, and reducing their representation in decisions made by these technologies (Giest and Samuels, 2020). Another important factor related to ethics in big data within organizations is the transparency of the data collection process, analysis, sharing, and access. For example, Manroop *et al.* (2024) explored the importance of transparency in the recruitment, training, and development processes and its impacts on trustworthiness, honesty, and integrity.

The ethics of big data in organizations can be approached by focusing on both organizational resources and the stakeholders involved in managing big data. Barney (1986) introduced the resource-based view model (RBV), which has had a significant impact on both theoretical and empirical research in the field of management. RBV is frequently cited and has influenced a wide range of academic disciplines. According to the RBV, organizations can gain a competitive advantage by leveraging resources that are valuable, rare, inimitable, and non-substitutable. Apart from organizational resources, ethics in big data has also been explored in relation to the stakeholders involved in its management. Randolph-Seng *et al.* (2024) proposed a theoretical framework focusing on big data disillusionment/enchantment, with company-level factors (adjustable features and default features) and individual-level factors. They extended previous theories on big data, ethics, and external stakeholders, such as customers, focusing on internal stakeholders, including employees.

3. Research method

In this research, we followed a predetermined research protocol (Kitchenham, 2004), aiming to examine and inspect the literature related to ethics in big data in the era of Industry 4.0. We utilized several synonyms and alternative terms to retrieve the dataset. We referred to the specific website (<https://www.powerthesaurus.org/>) to identify all possible synonyms for “ethics,” “big data,” and “Industry 4.0”. The authors reviewed the potential set of keywords, retained the most relevant ones, and confirmed them. By doing so, we ensured broad coverage of studies that align with the research topic. The corpus for the study was retrieved from the Scopus database. The query used in the Scopus database is presented in Figure 1. The research method is outlined in Figure 2. We followed the guidelines provided by several studies that conducted bibliometric analyses in various contexts (Carè and Cumming, 2024; Cherappurath *et al.*, 2024; Levidze, 2024; Lim *et al.*, 2024; Wang *et al.*, 2024; Yalcinkaya and Cinar Yucel, 2024; L. Zhu *et al.*, 2024). The review inspected all the articles up to 6 March 2023; which resulted in 277 documents in the time span of 1990–2023.

This research employed the Scopus database, an abstract and indexing electronic resource with full-manuscript links. Scopus includes nearly 20,000 peer-reviewed journals from well-recognized publishers such as Inderscience, Informa, Taylor & Francis, Springer, Emerald,

TITLE-ABS-KEY ((big AND data AND ethic* AND business) OR (big AND data AND ethic* AND industry) OR (big AND data AND ethic* AND industry 4.0) OR (big AND data AND ethic* AND organization*)) AND (LIMIT-TO (SRCTYPE , "j")) AND (LIMIT-TO (DOCTYPE , "ar")) AND (LIMIT-TO (SUBJAREA , "BUSI") OR LIMIT-TO (SUBJAREA , "ENGI") OR LIMIT-TO (SUBJAREA , "COMP") OR LIMIT-TO (SUBJAREA , "SOCI") OR LIMIT-TO (SUBJAREA , "DECI") OR LIMIT-TO (SUBJAREA , "ECON") OR LIMIT-TO (SUBJAREA , "ENVI")) AND (LIMIT-TO (LANGUAGE , "English"))

Figure 1. Research query. Source(s): Authors’ own creation/work

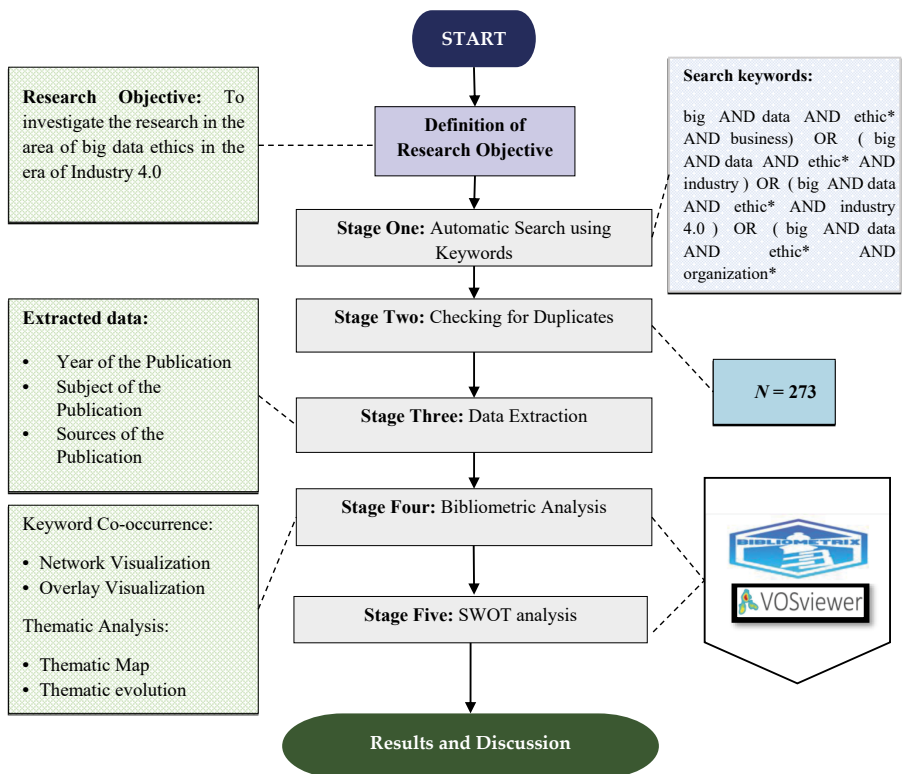


Figure 2. Research protocol. Source(s): Authors' own creation/work

and Elsevier (Yadav *et al.*, 2022). Because synchronizing bibliographic data from multiple databases can compromise the reliability of a bibliometric evaluation, only Scopus was used in this research. Scopus's unified layout and broad coverage make it a suitable resource for assessing research findings. Additionally, using a single database is common and considered standard in bibliometric analysis (Khuram *et al.*, 2023).

In the next stage, we performed an initial screening of the primary studies to identify duplicated and out-of-scope studies, resulting in 273 final studies for further analysis. The review of these studies was conducted by the first two authors to ensure the inclusion of only relevant studies. At this stage, the abstracts of all studies were read, and those not related to the research topic were excluded and removed from the Excel sheet used in the subsequent stage. The next step involved extracting data related to the distribution of the studies over time, their subjects, and sources. Following this, we used the VOSviewer tool to analyze the studies based on author keywords as the unit of analysis, focusing on co-occurrence analysis. Two types of visualizations were used: network and overlay diagrams. In the next stage of the research method, we conducted a SWOT analysis to examine the investigated studies, focusing on both external and internal variables, and considering the positive and negative aspects. It is important to note that other types of analysis, such as co-citation, citation, and bibliographic coupling, were not performed in this study, as they fall outside its scope. This study is more focused on the themes, developments, and progress within the research topic, as well as the internal and external factors related to it. Therefore, it integrates both bibliometrics and SWOT analysis to achieve the study's objectives.

4. Data extraction

4.1 The main information of the investigated literature

The initial screening of the included studies indicated that the research was published between 1990 and 2023. The total number of included sources is 217, while the total number of studies is 273. The number of single-authored documents is 78. The global co-authorship ratio is 24.91%, and the average number of co-authors per study is 3.02%. The total number of author keywords is 1,005. The average citations per document is 23.75%.

4.2 Temporal distribution of studies over the years

As presented in Figure 3, research on big data ethics in business has gained the interest of researchers, and we anticipate that the number of publications will continue to grow in the coming years. We observed that only one study was published in the years 2012, 2011, 2010, 2009, 2008, 2007, 2006, 2003, 1994, and 1990. Starting in 2014, the research topic gained more attention, with a varying number of publications each year. Research on this topic has grown between 1990 and 2022, with an annual growth rate of 6.07%. The highest number of publications occurred in 2021, with 50 studies, followed by 44 studies in both 2022 and 2020, and 37 studies in 2019.

4.3 Distribution of studies over the subject

The distribution of publications on the subject under study reflects its multidisciplinary nature, with diverse interrelated areas. We focused on the main topics as presented in the titles and abstracts of the studies. In Figure 4, we present the subjects with the highest frequency. Thirty studies primarily focused on business areas, while healthcare was the main subject of 26 studies. Education was the focus of 13 studies, and sustainability was the main area in another 13 studies. Data privacy was the primary topic in 12 studies. Topics related to auditing, industry, and society were each investigated in 10 studies.

4.4 Most relevant sources

To gain more insights into the sources of the explored studies, we used Biblioshiny to present the most relevant sources along with the number of publications for each source. As shown in Figure 5, the Journal of Business Ethics published the highest number of studies, with 9 studies. The next relevant sources are Sustainability and Technological Forecasting and Social Change, each with 6 studies. Both Ethics and Information Technology and Science and Engineering Ethics published four studies. Additionally, Big Data and Society, Journal of

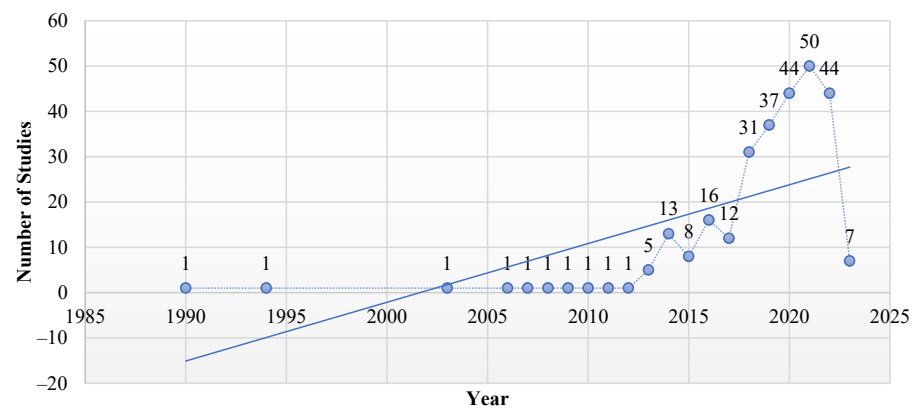


Figure 3. Temporal distribution of studies over the years. Source(s): Authors’ own creation/work



Figure 4. Distribution of Studies over the Subject. Source(s): Authors' own creation/work

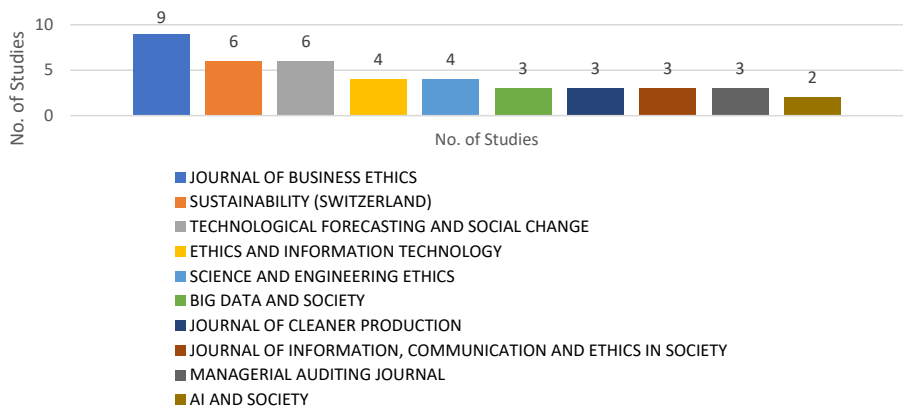


Figure 5. Most relevant studies. Source(s): Authors' own creation/work

Cleaner Production, and Journal of Information, Communication, and Ethics in Society each published three studies.

5. Bibliometric analysis

5.1 Keyword co-occurrences (VOSviewer)

Recently, bibliometric research has gained increasing attention from researchers in different disciplines, as it provides scholars with a holistic view of the research topic. After retrieving a piece of literature, bibliometric analysis can help visualize the trends, themes, development, and timespan of the published studies. In bibliometric studies, specific electronic resources are usually used to retrieve the studies, such as Scopus or Web of Science. Following that, the retrieved data is explored using different tools, aiming to analyze various types of relationships, such as co-occurrence, co-citations, bibliographic coupling, and co-authorship. The units of analysis are also diverse, including keywords, authors, organizations, and countries. In this research, the first stage of analysis was conducted using the VOSviewer tool. We chose to create a visual map based on the bibliographic data,

focusing on keyword co-occurrence. The tool examined the bibliographic database file downloaded from the Scopus database. The type of analysis is co-occurrence, and the unit of analysis is “Author Keywords”. The minimum number of keywords was set to 2, with 136 keywords from 1,005 met the condition. To eliminate redundancy from similar words, we utilized the VOSviewer tool, which allows us to input a file to replace similar terms with their synonyms. For each chosen word, the total strength of the co-occurrence links with other keywords was calculated, including only words with the highest total link strength. The analysis resulted in four clusters (Figure 6(a)): Cluster 1 (red), Cluster 2 (green), Cluster 3 (blue), and Cluster 4 (yellow). The first cluster includes words related to data ethics, such as ethics (48 occurrences), artificial intelligence (27 occurrences), machine learning (10 occurrences), and data governance (5 occurrences). In the second cluster, several keywords related to data are included, such as privacy (19 occurrences), confidentiality (2 occurrences), informed consent (2 occurrences), human rights (3 occurrences), and smart cities (5 occurrences). The third cluster includes keywords related to business and organizations, such as business ethics (7 occurrences), corporate social responsibility (8 occurrences), governance (7 occurrences), audit quality (2 occurrences), and job satisfaction (2 occurrences). In the fourth cluster, the keywords are more related to big data analytics techniques, such as big data (101 occurrences), data analytics (5 occurrences), content analysis (3 occurrences), data science (5 occurrences), and cloud computing (2 occurrences).

VOSviewer allows three types of visualizations: network, overlay, and density. In this study, we aimed to focus on the occurrences of the words (see Table A1 in Appendix), along with the development or progress of the research. As presented in Figure 6(b), the figure shows the progression of research over time. The research has focused on keywords related to data in health organizations, such as confidentiality, health records, and health data privacy. This result is not surprising, as indicated by Ryen (2023). In the last few decades, in response to misdemeanors and abuses in medical research, external and comprehensive protocols have been produced through the collaboration of different stakeholders. In 2018, the research focused on big data, data analytics, data science, and business analytics. In 2020, the research centered on artificial intelligence (Janssen *et al.*, 2020), data ethics, research ethics, and big data analytics (Bolton, 2020). After 2022, the research has focused on Industry 4.0 (Bag *et al.*, 2023), COVID-19 (Yallop *et al.*, 2023), consumers, and stakeholder theory (Bag *et al.*, 2023).

5.2 Thematic map (Biblioshiny)

The thematic maps and evolution of the keywords of the research are presented in Figure 7(a), Figure 7(b), Figure 8(a), and Figure 8(b). Figure 7(a) presents the alluvial diagram, which clarifies how different terms in a particular field display discontinuity, continuance, or transience. This allows the researcher to inspect how the keywords merge with others, disappear, emerge, or reappear over time (Leydesdorff and Schank, 2008). The diagram elaborates on the development of the concepts within the research area. The width of each line represents the level of significance between keywords in a particular period, while colors are used to reflect the diversity of topics in the research. As presented in the diagram, the evolution of the themes has followed different routes. Two main themes have contributed significantly to the development of the big data research theme: sustainable development and ethics. Research on privacy has been influenced by two core themes: big data and ethics. The theme of “big data” has also contributed to the advancement of several other research areas, including population statistics, technological development, and data analytics. Similarly, the theme of “ethics” has influenced multiple areas, such as privacy, information dissemination, human-based research, and behavioral research.

To support the interpretation of the alluvial diagram, a composite thematic map was generated (Figure 7(b)). The map is developed based on two measures: density, represented on the Y-axis, and centrality, shown on the X-axis. Density refers to the internal development of a

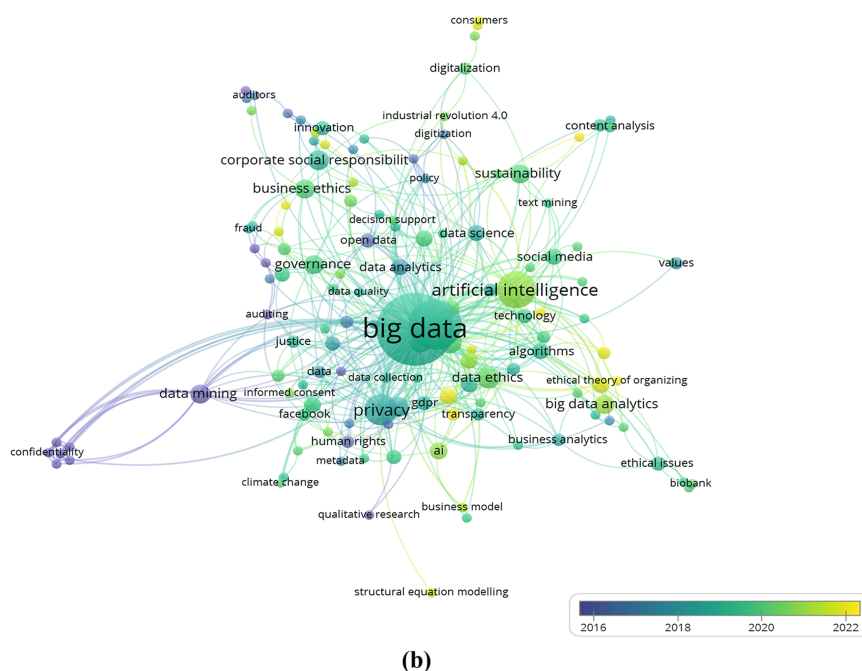
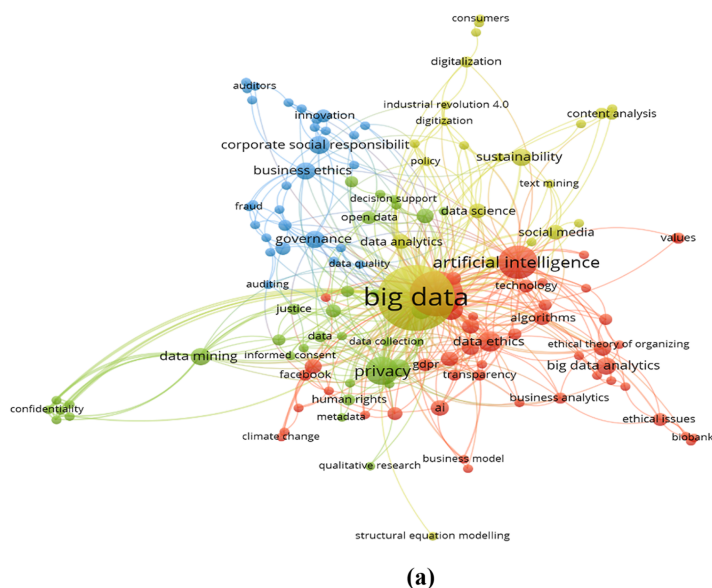


Figure 6. Keywords co-occurrence analysis. Source(s): Authors' own creation/work

keyword cluster (i.e. how developed the theme is), while centrality reflects its relevance or significance in the overall research network. These graphs provide valuable insight into theme development. In particular, they help the researcher understand how specific themes evolve into other thematic classifications, merge into new areas, or continue to grow independently.

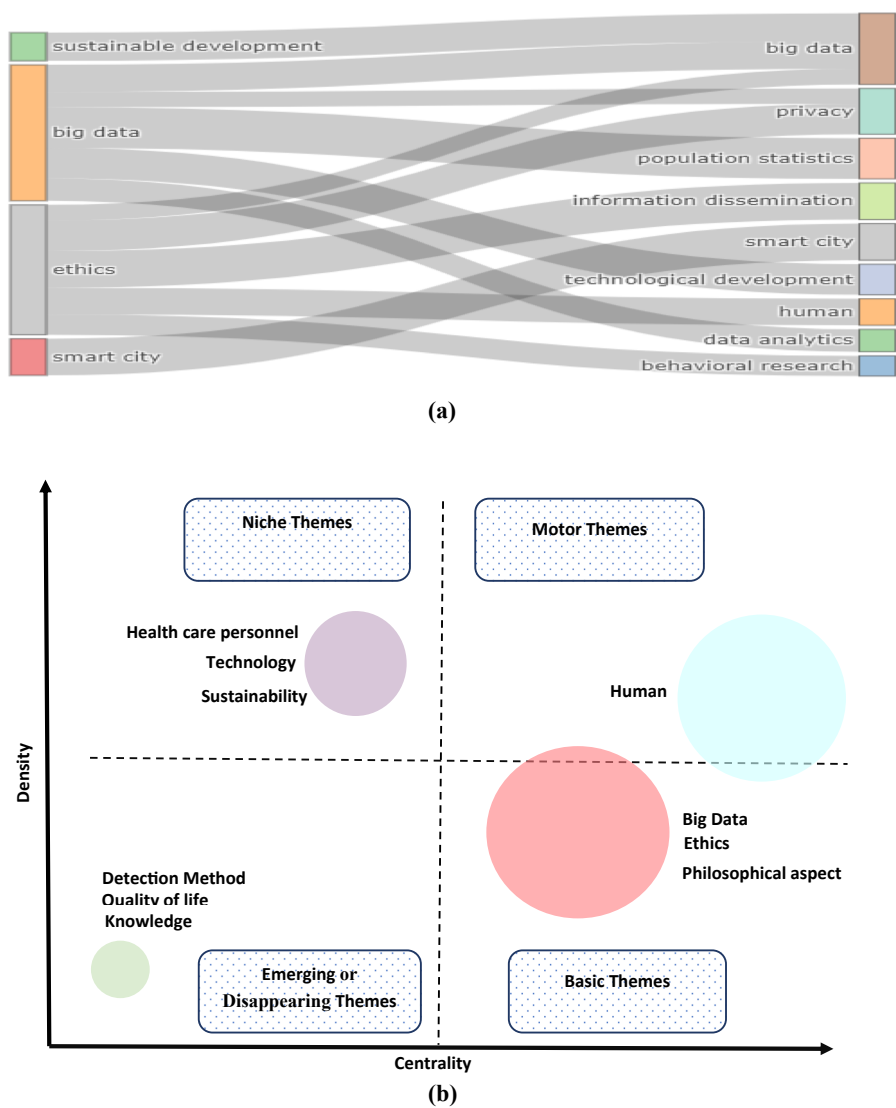


Figure 7. The results for (a) thematic evolution (1990–2020, 2021–2023) and (b) composite thematic map. Source(s): Authors’ own creation/work

The titles of the circles on the thematic map represent the most frequently occurring terms at a given time, while the size of each circle reflects the relative frequency of the word occurrences.

Thematic evolution in a research field can be visualized by dividing the entire timeline into several time slots. This method allows the progression of research themes to be traced and compared across different periods rather than focusing on the full timeline as a whole. Based on this, we selected 2020 as a cut-off year, and two thematic maps were created accordingly (Figure 8(a) and (b)).

As shown in Figure 8(a), motor themes represent influential keywords that demonstrate strong growth momentum. These include ethics, human, philosophical aspects, data analytics,

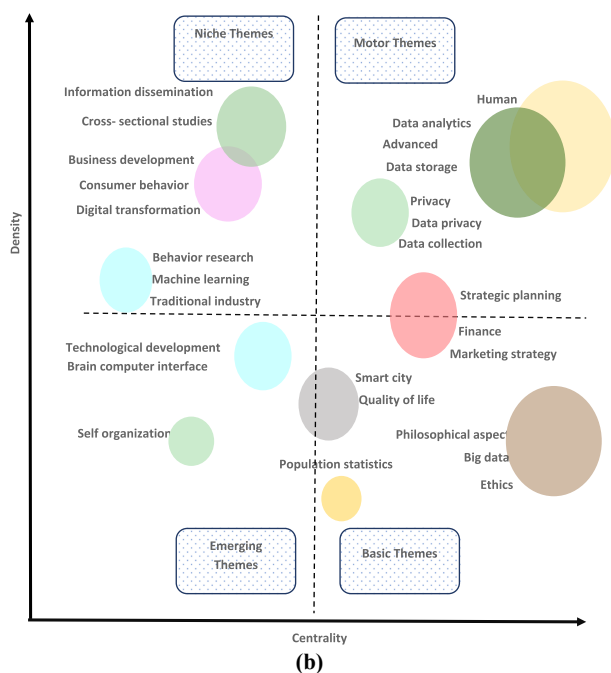
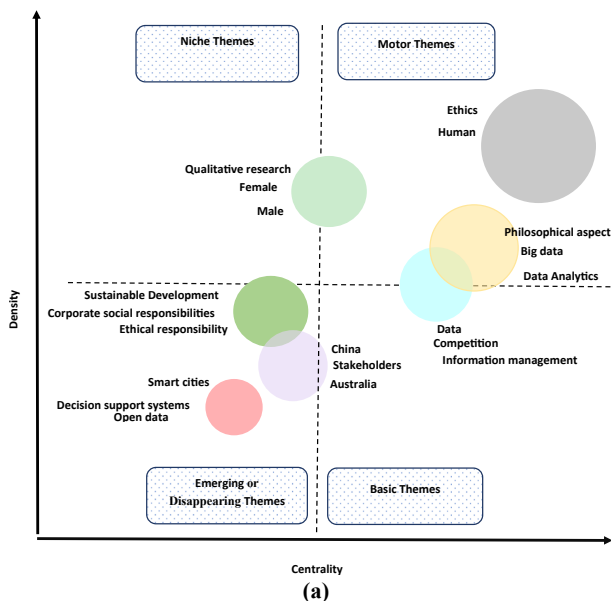


Figure 8. The results for (a) thematic map (time span 1990–2020) and (b) thematic map (time span 2020-up-to-date). Source(s): Authors' own creation/work

and big data. Niche keywords are highly developed but isolated, including terms such as qualitative study, female, and male. Emerging or disappearing keywords refer to those that have either newly appeared or faded from the literature, such as corporate social responsibility,

ethical responsibility, smart cities, decision support systems, and open data. Finally, basic keywords, which are considered foundational in the field, include terms like data competition and information management.

Focusing on [Figure 8\(b\)](#), motor themes in the second time slot (starting in 2021) increasingly reflect topics related to data analysis. Specifically, motor themes have expanded to include data analytics, advanced analytics, and data storage. Additional data-related keywords such as data privacy, and data collection also appear prominently within this cluster. These themes have been explored in several studies, including those by [Yallop et al. \(2023\)](#) and the study by [Burkhardt et al. \(2022\)](#). Conversely, philosophical aspects and big data, which were previously positioned as motor themes, have shifted to basic themes in this time slot, as seen in the work of [Bag et al. \(2023\)](#). Niche keywords during this period reflect a broader diversification and include terms such as information dissemination, cross-sectional studies, business development, consumer behavior, digital transformation, behavior research, machine learning, and traditional industry. Emerging keywords show a growing interest in advanced technologies, particularly in areas such as brain-computer interfaces, as discussed by [de Kerckhove \(2021\)](#), and technological development.

6. Strength, weakness, opportunities, and threats (SWOT) analysis

The growing digitalization of diverse social interactions has generated huge volumes of data, reaching zettabytes in volume on the World Wide Web ([Krotov and Johnson, 2022](#)). Big web data includes various forms such as videos, photos, posts, blogs, emails, and databases, stored as structured, unstructured, or semi-structured forms. This review has three main components: ethics, big data, and businesses. Specifically, we focus on ethics in relation to big data and its influence on business performance. The data science process in organizations aims to answer specific research questions, which are interlinked with the policies followed in the organization ([Provost and Fawcett, 2013](#)). The data analytics process involves several stages: collection, cleaning, analysis, and visualization ([Krotov and Johnson, 2022](#)). In all these stages, the ethics of data analysts, data scientists, and decision-makers play an important role in the organization's performance (See [Figure 9](#)). To gain the ultimate advantages of big data, it should be explored by considering both internal and external variables that influence its effective utilization. In this section, we aim to apply a SWOT analysis to investigate different aspects of the main components of the study.

SWOT analysis originated from an approach entitled the SOFT approach ([Kraus and Schwarz, 2007](#)). SWOT analysis is a powerful strategic method that offers a framework for evaluating key internal "strengths and weaknesses" as well as external "opportunities and threats" ([Longsheng et al., 2022](#)). In SWOT analysis, the researcher identifies internal and external factors and classifies them as strengths, weaknesses, opportunities, and threats. In this study, we referred to the insights driven by the bibliometric analysis to identify and classify these factors. Hence, in the following sections, we will examine these aspects through a SWOT analysis (see [Figure 10](#)).

6.1 Strength and weakness

- (1) *Data Quality*: It is important, when discussing ethics in big data, to focus on the quality of the data, as there is a strong relationship between the two. Data with low quality can lead to misleading decisions ([Ghasemaghahi and Calic, 2019](#)). The quality of data significantly affects research findings and their generalization, scientists' credibility, business outcomes, and political factors. Data quality is a critical factor in the outcome of any analytical framework or classification algorithm. Poor-quality data can cost organizations money and impact their decisions and plans ([Ghasemaghahi and Calic, 2019](#); [Grange et al., 2019](#)). Data quality is operationalized by referring to the area or field in which it is deployed; hence,

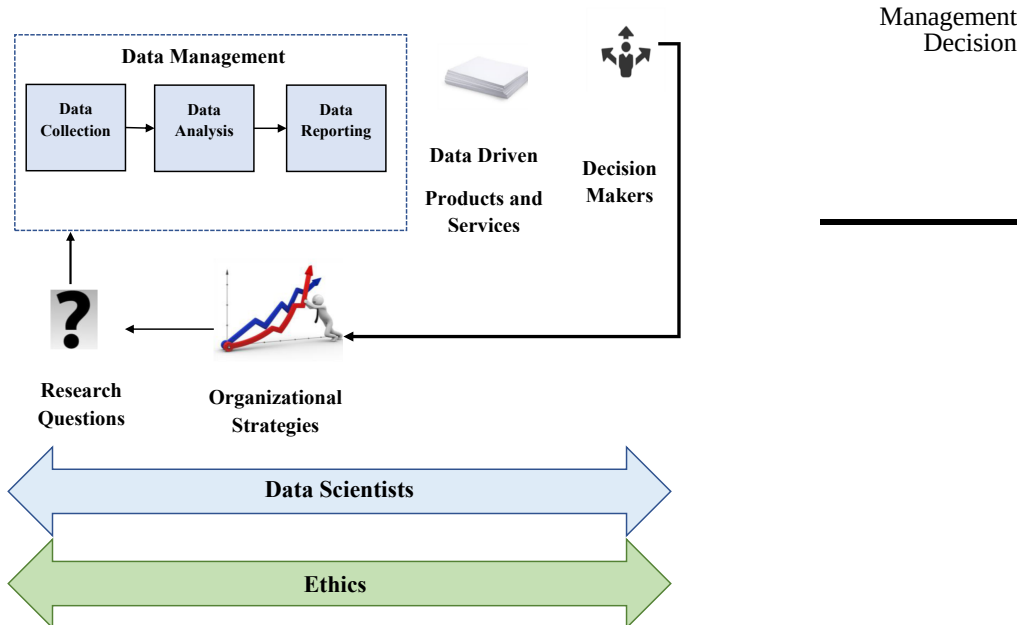


Figure 9. Ethics in the data science process. Source(s): Authors' own creation/work

multiple definitions of data quality have been proposed in the literature. Still, measuring the quality of data quantitatively is a complex task, as it entails different dimensions depending on the domain. Starting from the 90s, the appraisal of data quality has been subject to diverse and interrelated dimensions (Webb and Webb, 2004). Data quality is reflected by the degree to which it matches the owner's needs and the predetermined goals (McCord *et al.*, 2022). As indicated by Omri *et al.* (2021), intrinsic factors, contextual factors, and accessibility affect the level of data quality. Other studies have focused on three main aspects of data to explore data quality: completeness, volume, and accuracy. To reach the right decisions, both at the individual and organizational levels, the assessment of data has been subject to different metrics. Data diagnosticity, in which refined perceptions can be inferred from the data, depends entirely on the quality of the data (Ghasemaghaei and Calic, 2019). Data diagnosticity enables the organization to assess its current position in the market and examine the influential factors that affect its performance. The organization inspects the data to estimate its current position compared to its competitors, adding both tangible and intangible value to the organization (Erevelles *et al.*, 2016). In this context, several artificial intelligence approaches have been deployed, including deep learning, machine learning, and data mining. Ethics are related to these approaches in different ways. While data should meet a size requirement to provide significant outcomes, handling the data collection process, addressing missing values, determining the suitable sample size, and reporting these issues are considered important ethical concerns for data scientists. Data ethics is an emerging discipline that examines and analyzes moral problems associated with data throughout its life cycle (from collection to usage by the end customer), the techniques deployed when using data, and other related activities to establish strong ethical choices and behavior patterns (Floridi, 1999). Data ethics encompasses data processed by computational algorithms, which can inform decision-making and the procedures

Weaknesses Strengths Threats Opportunities		Data Quality	Information Dissemination	Data Privacy
		Misleading organizations decisions	Inequitable consideration of data	Breaching the privacy of social groups
		Impact Scientists' credibility	Unethical exposure to data	Endanger the organizations' image
		Data diagnostic	Development of research	Implementation of the Privacy by Design
		Establish strong ethical options	Goods and services advancement	Recognition of participants tasks
		Data governance	Global demand for codes of ethics	Global training of big data ethics
		Business ethics is rarely taken seriously	Usurp ethics' discretionary nature	A threat of the absence of big data ethics
		Different types of obstacles	Focus on the negative sides in evaluating ethics	Lack of data ethics experts
		Present a framework with different solutions	Globalization of economic activities	Integrate the ethics in all training programs
		Includes issues of political legitimacy	Global recognition of ethics in data science	Improved educational possibilities

Figure 10. Summary of the SWOT analysis. Source(s): Authors' own creation/work

for building and running such systems. An example of this is the algorithm used for data analysis, which should be deployed with consideration for algorithm ethics. The growing complexity and autonomy of algorithms pose significant obstacles in algorithm ethics, such as the moral responsibility and accountability of algorithm developers regarding the reporting of unpredicted and unwanted outcomes (Stefanizzi, 2022).

- (2) *Information Dissemination*: Information dissemination is explored as both a strength and a weakness, as we focus on the possible positive and negative aspects of data dissemination within the organization. Information dissemination has been explored in the literature in diverse contexts. One important area in the investigated literature is data sharing in the medical context. There are two perceptions of data sharing in the medical field: (1) the advancement of goods and services, and (2) making the right and ethical choice in terms of data sharing. In the context of data sharing in the medical field, making medical data accessible is a positive step. Recognizing the possibilities of data sharing and enabling people to cooperate and contribute data to research is a valuable aspect of the medical field. However, several challenges in this context should be carefully addressed, as data-sharing efforts might have negative consequences. Efforts to advance goods and services by making health-related data available can result in restrictions in the realm of what is right. Examples of the negative aspects include inequitable consideration of data from various sources and communities, and flaws in dissemination that might lead to exposure of medical data.

In the surveyed literature, a study by [Hummel and Braun \(2020\)](#) addressed these issues and suggested that governance is critical for preventing and mitigating injustice in data dissemination. It will also enable individuals to share data while allowing them to retract it if necessary or desired. Moving toward public-oriented ethics from research-oriented ethics raises several aspects that need to be considered, particularly the advantages and obstacles of data dissemination, focusing on the decision-making of legitimate choices in this context. Diverse factors can lead to the positive impacts of data sharing, including accountability, trust, equity, and proportionality ([Ballantyne, 2019](#)).

- (3) *Data privacy*: Data privacy is also closely interlinked with ethical perspectives in various areas of research. This is justified by the increasing privacy concerns that have become more prominent in the big data era, necessitating greater organizational attention ([Hong et al., 2021](#)). Concerns about data privacy frequently obstruct the acceptance and use of big data analytics, which has implications for digitalization and, ultimately, long-term sustainability. These issues were discussed in relation to informed consent and the legal duties of decision-makers by [Burkhardt et al. \(2022\)](#). Indeed, businesses that take advantage of the economic gains that big data promises but struggle to properly address privacy concerns might face severe consequences that will endanger their public image, capabilities, and overall competitive advantage ([Hajli et al., 2021](#)). To tackle these issues, the notion of privacy by design (PbD) is growing rapidly ([Alkhariji et al., 2023](#)), and it is a major issue for general data protection regulation ([Andrew and Baker, 2021](#)). PbD entails incorporating privacy laws into the design, implementation, and operation of computer systems, along with the participant's unrestricted and particular consent needed for the gathering, utilization, or dissemination of their information. One of the most urgent aspects in this area is the recognition of participants using methods for obtaining data-driven information, the connecting and incorporation of data sources, and the threat of breaching not only privacy rights but also the "privacy of social groups or communities", which could result in group discrimination based on, for instance, ethnic background or gender.
- (4) *Ethical Decision Making*: Ethical decision-making in organizations aims to assess and select alternatives in a way that aligns with ethical frameworks and values. It involves "choosing the right thing to do"; it is not about making clear-cut judgments of right and wrong ([Schwartz, 2016](#)). In many contexts, ethical decision-making is about deciding on the "lesser of two evils" and striking a balance between competing ideas and structures ([Power, Heavin and O'Connor, 2021](#)). Some technological choices require prioritizing conflicting ethical standards and ideals. Additionally, it is often necessary to rationalize the decision. For instance, one must carefully assess whether the legality or permissibility of a practice automatically makes it ethical. The utilization of big data in organizational decision-making, with a focus on its ethical implications, has been explored in the literature, highlighting both its advantages and potential shortcomings. For example, the application of big data analytics in human resource management offers numerous benefits for workforce management ([Manroop et al., 2024](#)). However, it is important to evaluate how organizational surveillance and other algorithmic biases, such as control, coercion, and profiling, can put workers at significant risk when big data analytics are used ([Manroop et al., 2024](#)). In another context, big data adoption enables organizations in industrial sectors to identify manufacturing process issues and quickly locate impacted customers. Still, despite its potential, big data is frequently employed in these sectors without consideration of its practical applications or alignment with observable corporate objectives and goals ([Polding and Eizaguirre Dieguez, 2021](#)). Furthermore, the degree to which big data

6.2 Opportunities and threats

- (1) *Data governance*: Data governance is a crucial component in helping organizations leverage and protect data (Vial, 2023). With a few notable exceptions, business ethics is rarely given serious attention in the boardroom unless prompted by a crisis. According to the literature, once an effort is made to create and disseminate a “code of ethics”, various obstacles are encountered by stakeholders (Webley, 2001). It is necessary to have a thorough understanding of data ethics in the establishment of businesses because data-based value creation may involve various routes. It is worth noting that a broader ethical perspective, focused on larger justice issues and addressing the underlying causes of injustice, could be adopted instead of focusing solely on individual moral issues related to data. Complying with data ethics in businesses does not imply that ethics must establish new norms in this sense. However, when data-driven procedures are engaged, ethical issues acquire novel approaches. As a result, complying with abstract ethical values and principles alone will likely not be sufficient for appropriate data governance in businesses. To begin, it is essential to comprehend where and how ethical difficulties arise and where organizations’ decision-makers may need to intervene to avoid ethically challenging types of data-driven value creation. Similarly, it is important to investigate which ethical issues decision-makers must consider, depending on the specific implementations that are incorporated. As a result, when it comes to governing database procedures in businesses, one solution does not fit all. Moreover, this value creation takes place in a public setting, includes issues of political legitimacy, and thus has an undeniably political component, which is determined by the specific data-driven processes that are implemented (König, 2021).
- (2) *Global demand for codes of ethics*: Given its value in supplementing scarce national statistics and supporting data-driven strategies, particularly during emergencies; international development and humanitarian organizations are increasingly advocating for digital data to be treated as a public good. In response, a “responsible data” movement has emerged to develop guidelines and frameworks for establishing ethical principles in data sharing. However, this initiative is facing challenges, as those holding critical data; especially mobile network operators; are often reluctant to share information obtained from low- and middle-income regions through intermediaries (Vayena and Tasioulas, 2016). Nonetheless, the general public continues to call for greater openness and accountability in both corporate and public spheres concerning big data. Furthermore, there is growing support for global recognition of the importance of fundamental human rights and environmental standards, which can be reinforced through ethical data practices. The application of ethical principles in big data has the potential to guide economic activities in business toward a more global outlook. However, on the downside, the discretionary nature of ethics may gradually be overtaken by strict regulations and legal frameworks. Additionally, pressure to measure ethical performance may emphasize negative shortcomings rather than highlighting positive progress.
- (3) *The need for global training*: The availability and affordability of educational initiatives have increased knowledge and awareness of the decision-making process within organizations. Such initiatives are particularly important in the context of the ethics of big data, as humans play a central role in implementing these ethics.

Organizations, accordingly, should adhere to these ethics at all stages, from data collection to decision-making (Zulkarnain *et al.*, 2021). This knowledge and awareness will increase the desire to become more actively involved in data sharing and transparency among different stakeholders. An overall absence of big data ethics topics in company training programs poses a vital threat to maintaining high moral standards at both employee and management levels. The best practice to address this threat is to include the aspect of ethical consciousness in all training courses that involve all stakeholders. Data scientists' ability to recognize and handle ethical concerns in data science has become increasingly important. Addressing current ethical problems in cooperative and multidisciplinary ways will facilitate data scientists' efforts by enabling the involvement of experts in sociological disciplines and improving researchers' capacity for critical thinking about the ethics of their tasks.

7. Discussion and research implications

Previous literature shed light on several practical and theoretical implications, which we will discuss broadly. Specifically, we will discuss the investigated literature and highlight implications referring to the theory, context, characteristics, and methodology (TCCM) framework. The TCCM framework focuses on the adopted theories, contexts or areas in which the research was carried out, interactions among the main factors, and techniques or methods of application (Ye, 2024).

The research offers a theoretical contribution by emphasizing the gap in the literature regarding the adoption of normative ethics and metaethics lenses in previous literature. While the majority of studies have focused on providing practical solutions to address several ethical risks linked to big data adoption in organizations, society, and governments, there is a call for a more concise understanding of concepts, definitions, and descriptions of ethics in the big data era. This is essential with the emergence of several conflicts within the context of big data adoption (e.g. privacy protection vs decision-making efficiency). There is also a need to understand the social norms regarding the adoption of big data analytics capabilities on both organizational and social levels.

Focusing on the practical implications, the review of the literature has stressed the importance of ethics in big data across many types of businesses. Tourism was among the main areas that gained the interest of researchers. Big data analytics offer many opportunities to analyze travelers' data and aid in the decision-making process (Yallop *et al.*, 2023), particularly during the COVID-19 crisis. Big data provides significant opportunities for marketing destinations. According to Zhu (2022), utilizing big data enables the analysis of customer data from multiple dimensions, facilitates the examination of customers' daily behavior patterns, helps identify customers' needs, and enables the creation of precise customer profiles. However, big data raised important concerns related to security, privacy, and data breaches. Accordingly, researchers have focused on addressing these concerns and providing frameworks and solutions.

Driven by the advancement of technology, the concept of privacy has changed over time, evolving from control over one's data to the protection of private information (Brodny and Tutak, 2025). Considering the enormous volumes of data that businesses, governments, and other organizations gather, store, and analyze; securing personal data has now become the main concern of organizations. According to Brodny and Tutak (2025), personal information that is gathered through a variety of procedures, when disclosed, may violate people's privacy or affect particular social and minority groups. Concerns regarding data privacy and the obligations associated with its management are pertinent in today's data-driven society. Yallop *et al.* (2023) proposed a framework that incorporates an ethics-driven approach, encompassing crucial aspects related to privacy, ethics, and the fair exchange of travelers' data. Kalvet *et al.* (2020)

explored the ethical concerns related to big data implementation in the tourism industry. The study highlighted various ethical concerns associated with big data utilized from social media and the web for analyzing tourism flows, visitor usage, and perceptions of tourism attractions. Focusing on industry, big data enables real-time customization of products according to [Tien \(2020\)](#). However, there is a need to address the moral, ethical, security, and employment issues related to both mass and real-time customization.

Big data has been explored broadly in several studies within the context of healthcare, offering several practical contributions. [Kaplan \(2015\)](#) pointed out that public expectations and privacy issues must be considered in ethical and policy studies. The authors argued that health data de-identification is often inadequate for individual data protection. The study also highlighted the conflict between data sharing and privacy concerns. While data sharing improves patient care and safety, privacy concerns could lead people to withhold information, thus compromising the validity of the data. Additionally, [Muller et al. \(2022\)](#) found a link between positive attitudes toward health data sharing and the need for personal decision-making, along with strategies to improve control at the collective level of governance. The risks associated with big data in healthcare call for the development of Big Data Ethics, integrating concepts such as privacy of information, protecting confidentiality when sharing private data, providing transparency, and protecting identity ([Stylianou and Talias, 2016](#)). Local initiatives should include proactive participation from physicians in societal and governmental policy discussions to address issues related to big data adoption and confront the new ethical challenges emerging from the evolving dynamics of healthcare ([Brill et al., 2019](#)).

The importance of meeting Sustainable Development Goals (SDGs) has been linked to advancements in communication and information technologies ([Bag et al., 2023](#)). The impact of big data analytics capabilities is evident in the context of sustainability initiatives, offering practical contributions to different stakeholders. The presence of sustainability is reflected in a comprehensive environment that encourages a transition from reactive to proactive analytics, aiming to generate value-driven information focused on water, waste, and energy issues ([Iacovidou et al., 2017](#)). Big data ethics concerning sustainability presence can be expressed through practices such as de-identifying information, safeguarding data from external threats, ensuring rightful ownership, and enabling evidence-based decision-making ([Ongsakul and Sen, 2018](#)). [Seele \(2017\)](#) discussed the possibility of adapting big data algorithms created for “predictive policing” to enhance corporate sustainability. Predictive sustainability control involves using analytical techniques to pinpoint subjects for collaborative discussion, oversight, and intervention, aiming to prevent future harm associated with environmental, social, and governance issues, address past threats, and identify potential actors and corporations engaged in unsustainable practices, along with their stakeholders.

Integrating sustainability-focused data practices within strategic data management practices highlights the value of big data, as a driver of sustainable development and an enabler for competitive success. Big data is now considered one of the most valuable resources for organizations. Building on the knowledge-based view ([Chowdhury et al., 2022](#)) and resource-based view ([Barney, 1986](#)), knowledge is a strategic asset of the organization and has a significant impact on its competitive advantage. Still, recognizing the importance of this asset, it is noted that knowledge is not easy to understand, evolves continuously, and is embedded with other assets of the organization, highlighting the importance of big data analytics. Previous literature has explored the role of knowledge on organizations’ performance, with an emphasis on the role of big data. For instance, [Feng et al. \(2024\)](#) explored big data as an important knowledge-based asset, focusing on its impact on the organization’s performance in terms of increased returns and value creation. Big data, as a valuable asset, when utilized effectively, can help organizations manage limited resources, thereby supporting sustainability goals ([Nilashi et al., 2023](#)).

Ethics in the big data era have been addressed across several dimensions (e.g. individuals, research, and policy). Literature has explored the ethical concerns of individuals and customers. At the individual level, the literature stresses that the needs and requirements of

users should be carefully considered. For instance, [Mogaji et al. \(2020\)](#) focused on vulnerable customers in the age of AI and highlighted the importance of human interaction to enhance customers' experience with financial providers. At the research level, there is a call for providing interpretative frameworks that address all ethical concerns. The literature has highlighted the importance of research in big data and suggested several practical guidelines to enhance the research outcomes. For instance, according to [Winickoff et al. \(2016\)](#), to improve the relevance and quality of health-based research in big data, there is a call for more efficient public engagement. On a policy level, both regulations and policies from authorities and organizations should be revised and updated to ensure control over data acquisition, analysis, and sharing of users' data. For instance, [Olesen \(2019\)](#) focused on the politics of whistleblowing and discussed three key aspects regarding whistleblowing in the big data era: its nature as a political action, its historical and social roots, and the practical impact of digitalization and big data.

The research stressed the importance of understanding the ethical responsibilities within organizations in the age of big data. Ethics can be approached through deontological approaches, teleological approaches, and virtue ethicists. In the context of big data, deontological approaches focus on formalizing ethical theories and principles, viewing ethics as closely related to regulations and policies ([van Maanen, 2022](#)). Teleological approaches aim to forecast or estimate the hazards that emerging technologies related to big data adoption may pose, and, based on these assessments, determine if and how a technology should be deployed. Finally, virtue ethicists focus on how practitioners apply moral principles throughout the adoption process ([van Maanen, 2022](#)). These ethical frameworks are absolutely important when evaluating the particular ethical issues related to big data adoption in organizations. Adopting big data by organizations without customers' permission is among the critical risks, as discussed in several studies in the literature. [Rautdesai et al. \(2019\)](#) stressed the need to implement self-regulating operating standards and practices by businesses and organizations to safeguard customers' privacy. Other studies stressed the need for responsible governance through regulatory policy to protect data privacy ([Leonard, 2014](#)). [Ulltveit-Moe \(2014\)](#) suggested a roadmap of privacy-enhanced intrusion detection systems, beginning with a privacy alarm management scheme and expanding to fully support privacy-focused operations.

Big data, depending on how it is used, can either progress or compromise human rights in line with teleological and deontological perspectives. Moving beyond privacy, human rights have been explored as a significant factor in the adoption of big data in organizations. Big data can help address several risks to human rights, facilitating safety, improving public health, and addressing violence. However, adopting big data may also raise several risks to human rights, including algorithmic bias, potential discrimination, manipulation and coercion, lack of transparency, and power imbalances ([Nersessian, 2018](#)). Actors in organizations are required by law to uphold human rights in their business operations, which inevitably involve big data-related activities ([Nersessian, 2018](#)).

The investigated literature has adopted several methods to explore the ethics in the big data era, including questionnaires ([Bag et al., 2023](#); [Nawawi and Ahmad Saiful Azlin Puteh, 2018](#); [Shafiq et al., 2020](#)), review of literature, theoretical explorations, and developing of conceptual frameworks ([Kernaghan, 2014](#); [Khder, 2021](#); [Ongsakul and Sen, 2018](#); [Yallop and Seraphin, 2020](#); [Yallop et al., 2023](#)), case study-based researches ([Berbenni-Rehm, 2023](#); [Ryan et al., 2020](#)), and Delphi-based research ([Asadi Someh et al., 2019](#)).

8. Conclusion

Ethics has a significant impact on organizational decisions at all levels. However, with a few notable exceptions, the ethical implications of big data in business organizations are rarely taken seriously. The primary goal of this research was to map, visualize, and analyze the evolution of research in the field of big data ethics, with a general focus on the industry and a

specific emphasis on organizational performance and business value. To achieve this goal, we searched the Scopus database for articles related to big data ethics in business. The method adopted, combining bibliometric and SWOT analyses of existing literature, offers decision-makers valuable insights and a comprehensive understanding of the current state of knowledge from multiple perspectives. The bibliometric analysis employed visual representations to highlight key elements and their interrelationships, enabling a clearer understanding of how previous studies have addressed the topic. The SWOT analysis, on the other hand, assessed the strengths, weaknesses, opportunities, and threats influencing the issue. Together, these approaches provide a more holistic and in-depth understanding of the subject under investigation.

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Table A1. Keywords co-occurrences and total link strengths

Keyword	Occurrences	Total link strength	Keyword	Occurrences	Total link strength
Big data	101	230	Tourism	4	10
Ethics	48	108	Accountability	3	11
Artificial intelligence	27	60	Automation	3	11
Privacy	19	55	Business Analytics	3	7
Machine learning	10	32	Content analysis	3	4
Data ethics	9	23	Data	3	8
Corporate social responsibility	8	9	Digital divide	3	10
Surveillance	8	26	Digital economy	3	4
Analytics	7	16	Digitalization	3	5
Big data analytics	7	7	Emerging technologies	3	5
Business ethics	7	15	Facebook	3	10
Data mining	7	31	human rights	3	8
Governance	7	20	Internet	3	9
Sustainability	7	9	Internet of things	3	10
AI	6	12	Justice	3	8
Algorithms	5	18	Management	3	4
COVID-19	5	10	Personal data	3	8
Data Analytics	5	17	Stakeholder theory	3	7
Data governance	5	15	Transparency	3	6
Data Science	5	18	Values	3	2
Datafication	5	19	Administrative data	2	6
Industry 4.0	5	10	Audit quality	2	2
Research ethics	5	11	Auditing	2	6
Smart cities	5	12	Auditor independence	2	2
Social media	5	14	Auditors	2	4
Big data ethics	4	6	Biobank	2	4
Data protection	4	7	Business intelligence	2	5
Entrepreneurship	4	10	Business model	2	6
Ethical issues	4	8	Change	2	4
GDPR	4	15	China	2	2
Innovation	4	8	Climate change	2	2
Open data	4	12	Cloud computing	2	8
Technology	4	7	Code of conduct	2	4

Source(s): Authors' own creation/work

References

- Abouelmehdi, K., Beni-Hessane, A. and Khaloufi, H. (2018), "Big healthcare data: preserving security and privacy", *Journal of Big Data*, Vol. 5 No. 1, pp. 1-18, doi: [10.1186/s40537-017-0110-7](https://doi.org/10.1186/s40537-017-0110-7).
- Alkhariji, L., De, S., Rana, O. and Perera, C. (2023), "Semantics-based privacy by design for Internet of Things applications", *Future Generation Computer Systems*, Vol. 138, pp. 280-295, doi: [10.1016/j.future.2022.08.013](https://doi.org/10.1016/j.future.2022.08.013).
- Alshawhi, S., Missi, F. and Irani, Z. (2011), "Organisational, technical and data quality factors in CRM adoption—SMEs perspective", *Industrial Marketing Management*, Vol. 40 No. 3, pp. 376-383, doi: [10.1016/j.indmarman.2010.08.006](https://doi.org/10.1016/j.indmarman.2010.08.006).

- Ananny, M. (2016), "Toward an ethics of algorithms: convening, observation, probability, and timeliness", *Science, Technology and Human Values*, Vol. 41 No. 1, pp. 93-117, doi: [10.1177/0162243915606523](https://doi.org/10.1177/0162243915606523).
- Andrew, J. and Baker, M. (2021), "The general data protection regulation in the age of surveillance capitalism", *Journal of Business Ethics*, Vol. 168 No. 3, pp. 565-578, doi: [10.1007/s10551-019-04239-z](https://doi.org/10.1007/s10551-019-04239-z).
- Asadi Someh, I., Davern, M., Breidbach, C. and Shanks, G. (2019), "Ethical issues in big data analytics: a stakeholder perspective", *Communications of the Association for Information Systems*, Vol. 44, pp. 718-747, doi: [10.17705/1CAIS.04434](https://doi.org/10.17705/1CAIS.04434).
- Bag, S., Rahman, M.S., Srivastava, G., Shore, A. and Ram, P. (2023), "Examining the role of virtue ethics and big data in enhancing viable, sustainable, and digital supply chain performance", *Technological Forecasting and Social Change*, Vol. 186, 122154, doi: [10.1016/j.techfore.2022.122154](https://doi.org/10.1016/j.techfore.2022.122154).
- Ballantyne, A. (2019), "Adjusting the focus: a public health ethics approach to data research", *Bioethics*, Vol. 33 No. 3, pp. 357-366, doi: [10.1111/bioe.12551](https://doi.org/10.1111/bioe.12551).
- Barney, J.B. (1986), "Organizational culture: can it be a source of sustained competitive advantage?", *Academy of Management Review*, Vol. 11 No. 3, pp. 656-665, doi: [10.2307/258317](https://doi.org/10.2307/258317).
- Berbenni-Rehm, C. (2023), "Evidence-based AI, ethics and the circular economy of knowledge", *AI and Society*, Vol. 38 No. 2, pp. 889-895, doi: [10.1007/s00146-022-01581-1](https://doi.org/10.1007/s00146-022-01581-1).
- Bolton, R.N. (2020), "Commentary: future directions of the service discipline", *Journal of Services Marketing*, Vol. 34 No. 3, pp. 279-289, doi: [10.1108/jsm-02-2020-0067](https://doi.org/10.1108/jsm-02-2020-0067).
- Brill, S.B., Moss, K.O. and Prater, L. (2019), "Transformation of the doctor–patient relationship: big data, accountable care, and predictive health analytics", *HEC Forum*, Vol. 31 No. 4, pp. 261-282, doi: [10.1007/s10730-019-09377-5](https://doi.org/10.1007/s10730-019-09377-5).
- Brodny, J. and Tutak, M. (2025), "Stakeholder interactions and ethical imperatives in big data and AI development", *Journal of Open Innovation: Technology, Market, and Complexity*, Vol. 11 No. 1, 100491, doi: [10.1016/j.joitmc.2025.100491](https://doi.org/10.1016/j.joitmc.2025.100491).
- Burkhardt, G., Boy, F., Doneddu, D. and Hajli, N. (2022), "Privacy behaviour: a model for online informed consent", *Journal of Business Ethics*, Vol. 186, pp. 1-19, doi: [10.1007/s10551-022-05202-1](https://doi.org/10.1007/s10551-022-05202-1).
- Caballero, I., Gualo, F., Rodríguez, M. and Piattini, M. (2022), "BR4DQ: a methodology for grouping business rules for data quality evaluation", *Information Systems*, Vol. 109, 102058, doi: [10.1016/j.is.2022.102058](https://doi.org/10.1016/j.is.2022.102058).
- Carè, R. and Cumming, D. (2024), "Technology and automation in financial trading: a bibliometric review", *Research in International Business and Finance*, Vol. 71, 102471, doi: [10.1016/j.ribaf.2024.102471](https://doi.org/10.1016/j.ribaf.2024.102471).
- Cherappurath, N., Shamshadali, P., Elayaraja, M. and Ki, D.A.K. (2024), "Mapping the field: a bibliometric analysis of women's football research trends and future directions", *Apunts Sports Medicine*, Vol. 59 No. 223, 100448, doi: [10.1016/j.apunsm.2024.100448](https://doi.org/10.1016/j.apunsm.2024.100448).
- Chowdhury, S., Budhwar, P., Dey, P.K., Joel-Edgar, S. and Abadie, A. (2022), "AI-employee collaboration and business performance: integrating knowledge-based view, socio-technical systems and organisational socialisation framework", *Journal of Business Research*, Vol. 144, pp. 31-49, doi: [10.1016/j.jbusres.2022.01.069](https://doi.org/10.1016/j.jbusres.2022.01.069).
- Cuquet, M. and Fensel, A. (2018), "The societal impact of big data: a research roadmap for Europe", *Technology in Society*, Vol. 54, pp. 74-86, doi: [10.1016/j.techsoc.2018.03.005](https://doi.org/10.1016/j.techsoc.2018.03.005).
- Dattner, B., Chamorro-Premuzic, T., Buchband, R. and Schettler, L. (2019), "The legal and ethical implications of using AI in hiring", *Harvard Business Review*, available at: <https://hbr.org/2019/04/the-legal-and-ethical-implications-of-using-ai-in-hiring> (accessed 2 May 2025).
- de Kerckhove, D. (2021), "The personal digital twin, ethical considerations", *Philosophical Transactions of the Royal Society A*, Vol. 379 No. 2207, 20200367, doi: [10.1098/rsta.2020.0367](https://doi.org/10.1098/rsta.2020.0367).

- Di, D. (2023), "Ethical ambiguity and complexity: tech workers' perceptions of big data ethics in China and the US", *Information, Communication and Society*, Vol. 26 No. 5, pp. 1-17, doi: [10.1080/1369118x.2023.2166357](https://doi.org/10.1080/1369118x.2023.2166357).
- Eastwood, C., Klerkx, L., Ayre, M. and Dela Rue, B. (2019), "Managing socio-ethical challenges in the development of smart farming: from a fragmented to a comprehensive approach for responsible research and innovation", *Journal of Agricultural and Environmental Ethics*, Vol. 32 No. 5, pp. 741-768, doi: [10.1007/s10806-017-9704-5](https://doi.org/10.1007/s10806-017-9704-5).
- Egwuonwu, A., Mendy, J., Smart-Oruh, E. and Egwuonwu, A. (2024), "Drivers of big data analytics' adoption and implications of management decision-making on big data adoption and firms' financial and nonfinancial performance: evidence from Nigeria's manufacturing and service industries", *IEEE Transactions on Engineering Management*, Vol. 71, pp. 11907-11922, doi: [10.1109/TEM.2023.3321426](https://doi.org/10.1109/TEM.2023.3321426).
- Erevelles, S., Fukawa, N. and Swayne, L. (2016), "Big Data consumer analytics and the transformation of marketing", *Journal of Business Research*, Vol. 69 No. 2, pp. 897-904, doi: [10.1016/j.jbusres.2015.07.001](https://doi.org/10.1016/j.jbusres.2015.07.001).
- Fan, W. (2015), "Data quality: from theory to practice", *Acm Sigmod Record*, Vol. 44 No. 3, pp. 7-18, doi: [10.1145/2854006.2854008](https://doi.org/10.1145/2854006.2854008).
- Feng, N., Tu, S. and Guo, F. (2024), "Big-data analytics capability, value creation process, and collaboration innovation quality in manufacturing enterprises: a knowledge-based view", *Computers and Industrial Engineering*, Vol. 187, 109804, doi: [10.1016/j.cie.2023.109804](https://doi.org/10.1016/j.cie.2023.109804).
- Floridi, L. (1999), "Information ethics: on the philosophical foundation of computer ethics", *Ethics and Information Technology*, Vol. 1 No. 1, pp. 33-52, doi: [10.1023/a:1010018611096](https://doi.org/10.1023/a:1010018611096).
- Forgó, N., Händold, S., van den Hoven, J., Krügel, T., Lishchuk, I., Mahieu, R., Monreale, A., Pedreschi, D., Pratesi, F. and van Putten, D. (2021), "An ethico-legal framework for social data science", *International Journal of Data Science and Analytics*, Vol. 11 No. 4, pp. 377-390, doi: [10.1007/s41060-020-00211-7](https://doi.org/10.1007/s41060-020-00211-7).
- Ghasemaghaei, M. (2020), "Improving organizational performance through the use of big data", *Journal of Computer Information Systems*, Vol. 60 No. 5, pp. 395-408, doi: [10.1080/08874417.2018.1496805](https://doi.org/10.1080/08874417.2018.1496805).
- Ghasemaghaei, M. and Calic, G. (2019), "Can big data improve firm decision quality? The role of data quality and data diagnosticity", *Decision Support Systems*, Vol. 120, pp. 38-49, doi: [10.1016/j.dss.2019.03.008](https://doi.org/10.1016/j.dss.2019.03.008).
- Giest, S. and Samuels, A. (2020), "'For good measure': data gaps in a big data world", *Policy Sciences*, Vol. 53 No. 3, pp. 559-569, doi: [10.1007/s11077-020-09384-1](https://doi.org/10.1007/s11077-020-09384-1).
- Grange, C., Benbasat, I. and Burton-Jones, A. (2019), "With a little help from my friends: cultivating serendipity in online shopping environments", *Information and Management*, Vol. 56 No. 2, pp. 225-235, doi: [10.1016/j.im.2018.06.001](https://doi.org/10.1016/j.im.2018.06.001).
- Greene, T., Dhurandhar, A. and Shmueli, G. (2023), "Atomist or holist? A diagnosis and vision for more productive interdisciplinary AI ethics dialogue", *Patterns*, Vol. 4 No. 1, 100652, doi: [10.1016/j.patter.2022.100652](https://doi.org/10.1016/j.patter.2022.100652).
- Grodzinsky, F.S. (2017), "Why big data needs the virtues", in *Philosophy and Computing: Essays in Epistemology, Philosophy of Mind, Logic, and Ethics*, pp. 221-234.
- Grover, V., Chiang, R.H., Liang, T.-P. and Zhang, D. (2018), "Creating strategic business value from big data analytics: a research framework", *Journal of Management Information Systems*, Vol. 35 No. 2, pp. 388-423, doi: [10.1080/07421222.2018.1451951](https://doi.org/10.1080/07421222.2018.1451951).
- Hajli, N., Shirazi, F., Tajvidi, M. and Huda, N. (2021), "Towards an understanding of privacy management architecture in big data: an experimental research", *British Journal of Management*, Vol. 32 No. 2, pp. 548-565, doi: [10.1111/1467-8551.12427](https://doi.org/10.1111/1467-8551.12427).
- Herschel, R. and Miori, V.M. (2017), "Ethics & big data", *Technology in Society*, Vol. 49, pp. 31-36, doi: [10.1016/j.techsoc.2017.03.003](https://doi.org/10.1016/j.techsoc.2017.03.003).
- Holmes, W., Porayska-Pomsta, K., Holstein, K., Sutherland, E., Baker, T., Shum, S.B., Santos, O.C., Rodrigo, M.T., Cukurova, M., Koedinger, K.R. and Bittencourt, I.I. (2021), "Ethics of AI in

- education: towards a community-wide framework”, *International Journal of Artificial Intelligence in Education*, Vol. 32 No. 3, pp. 1-23, doi: [10.1007/s40593-021-00239-1](https://doi.org/10.1007/s40593-021-00239-1).
- Hong, W., Chan, F.K. and Thong, J.Y. (2021), “Drivers and inhibitors of internet privacy concern: a multidimensional development theory perspective”, *Journal of Business Ethics*, Vol. 168 No. 3, pp. 539-564, doi: [10.1007/s10551-019-04237-1](https://doi.org/10.1007/s10551-019-04237-1).
- Hummel, P. and Braun, M. (2020), “Just data? Solidarity and justice in data-driven medicine”, *Life Sciences, Society and Policy*, Vol. 16 No. 1, pp. 1-18, doi: [10.1186/s40504-020-00101-7](https://doi.org/10.1186/s40504-020-00101-7).
- Iacovidou, E., Velis, C.A., Purnell, P., Zwirner, O., Brown, A., Hahladakis, J., Millward-Hopkins, J. and Williams, P.T. (2017), “Metrics for optimising the multi-dimensional value of resources recovered from waste in a circular economy: a critical review”, *Journal of Cleaner Production*, Vol. 166, pp. 910-938, doi: [10.1016/j.jclepro.2017.07.100](https://doi.org/10.1016/j.jclepro.2017.07.100).
- Janssen, M., Brous, P., Estevez, E., Barbosa, L.S. and Janowski, T. (2020), “Data governance: organizing data for trustworthy artificial intelligence”, *Government Information Quarterly*, Vol. 37 No. 3, 101493, doi: [10.1016/j.giq.2020.101493](https://doi.org/10.1016/j.giq.2020.101493).
- Kalvet, T., Olesk, M., Tiits, M. and Raun, J. (2020), “Innovative tools for tourism and cultural tourism impact assessment”, *Sustainability*, Vol. 12 No. 18, p. 7470, doi: [10.3390/su12187470](https://doi.org/10.3390/su12187470), available at: <https://www.mdpi.com/2071-1050/12/18/7470>
- Kaplan, B. (2015), “Selling health data: de-identification, privacy, and speech”, *Cambridge Quarterly of Healthcare Ethics*, Vol. 24 No. 3, pp. 256-271, doi: [10.1017/s0963180114000589](https://doi.org/10.1017/s0963180114000589).
- Kernaghan, K. (2014), “Digital dilemmas: values, ethics and information technology”, *Canadian Public Administration*, Vol. 57 No. 2, pp. 295-317, doi: [10.1111/capa.12069](https://doi.org/10.1111/capa.12069).
- Khder, M.A. (2021), “Web scraping or web crawling: state of art, techniques, approaches and application”, *International Journal of Advances in Soft Computing and Its Applications*, Vol. 13 No. 3, pp. 145-168, doi: [10.15849/ijasca.211128.11](https://doi.org/10.15849/ijasca.211128.11).
- Khuram, S., Rehman, C.A., Nasir, N. and Elahi, N.S. (2023), “A bibliometric analysis of quality assurance in higher education institutions: implications for assessing university’s societal impact”, *Evaluation and Program Planning*, Vol. 99, 102319, doi: [10.1016/j.evalprogplan.2023.102319](https://doi.org/10.1016/j.evalprogplan.2023.102319).
- Kitchenham, B. (2004), *Procedures for Performing Systematic Reviews*, Vol. 33, Keele University, Keele, pp. 1-26.
- König, P.D. (2021), “Citizen-centered data governance in the smart city: from ethics to accountability”, *Sustainable Cities and Society*, Vol. 75, 103308, doi: [10.1016/j.scs.2021.103308](https://doi.org/10.1016/j.scs.2021.103308).
- Kraus, S. and Schwarz, E. (2007), “The role of pre-start-up planning in new small business”, *International Journal of Management and Enterprise Development*, Vol. 4 No. 1, pp. 1-17, doi: [10.1504/ijmed.2007.011452](https://doi.org/10.1504/ijmed.2007.011452).
- Krotov, V. and Johnson, L. (2022), “Big web data: challenges related to data, technology, legality, and ethics”, *Business Horizons*, Vol. 66 No. 4, pp. 481-491, doi: [10.1016/j.bushor.2022.10.001](https://doi.org/10.1016/j.bushor.2022.10.001).
- Krotov, V. and Johnson, L. (2023), “Big web data: challenges related to data, technology, legality, and ethics”, *Business Horizons*, Vol. 66 No. 4, pp. 481-491, doi: [10.1016/j.bushor.2022.10.001](https://doi.org/10.1016/j.bushor.2022.10.001).
- Kuc-Czarnecka, M. and Olczyk, M. (2020), “How ethics combine with big data: a bibliometric analysis”, *Humanities and Social Sciences Communications*, Vol. 7 No. 1, p. 137, doi: [10.1057/s41599-020-00638-0](https://doi.org/10.1057/s41599-020-00638-0).
- Leonard, P. (2014), “Customer data analytics: privacy settings for ‘big data’ business”, *International Data Privacy Law*, Vol. 4 No. 1, pp. 53-68, doi: [10.1093/idpl/ipt032](https://doi.org/10.1093/idpl/ipt032).
- Lerman, J. (2013), “Big data and its exclusions”, *Stanford Law Review Online*, Vol. 66, p. 55.
- Levidze, M. (2024), “Mapping the research landscape: a bibliometric analysis of e-learning during the COVID-19 pandemic”, *Heliyon*, Vol. 10 No. 13, e33875, doi: [10.1016/j.heliyon.2024.e33875](https://doi.org/10.1016/j.heliyon.2024.e33875).
- Leydesdorff, L. and Schank, T. (2008), “Dynamic animations of journal maps: indicators of structural changes and interdisciplinary developments”, *Journal of the American Society for Information Science and Technology*, Vol. 59 No. 11, pp. 1810-1818, doi: [10.1002/asi.20891](https://doi.org/10.1002/asi.20891).

- Lim, W.M., Kumar, S. and Donthu, N. (2024), "How to combine and clean bibliometric data and use bibliometric tools synergistically: guidelines using metaverse research", *Journal of Business Research*, Vol. 182, 114760, doi: [10.1016/j.jbusres.2024.114760](https://doi.org/10.1016/j.jbusres.2024.114760).
- Liu, H., Lu, F., Shi, B., Hu, Y. and Li, M. (2023), "Big data and supply chain resilience: role of decision-making technology", *Management Decision*, Vol. 61 No. 9, pp. 2792-2808, doi: [10.1108/md-12-2021-1624](https://doi.org/10.1108/md-12-2021-1624).
- Longsheng, C., Ali Shah, S.A., Solangi, Y.A., Ahmad, M. and Ali, S. (2022), "An integrated SWOT-multi-criteria analysis of implementing sustainable waste-to-energy in Pakistan", *Renewable Energy*, Vol. 195, pp. 1438-1453, doi: [10.1016/j.renene.2022.06.112](https://doi.org/10.1016/j.renene.2022.06.112).
- Manroop, L., Malik, A. and Milner, M. (2024), "The ethical implications of big data in human resource management", *Human Resource Management Review*, Vol. 34 No. 2, 101012, doi: [10.1016/j.hrmr.2024.101012](https://doi.org/10.1016/j.hrmr.2024.101012).
- Mark, R. (2019), "Ethics of using AI and big data in agriculture: the case of a large agriculture multinational", *The ORBIT Journal*, Vol. 2 No. 2, pp. 1-27, doi: [10.29297/orbit.v2i2.109](https://doi.org/10.29297/orbit.v2i2.109).
- Mark, R. and Anya, G. (2019), "Ethics of using smart city AI and big data: the case of four large European cities", *The ORBIT Journal*, Vol. 2 No. 2, pp. 1-36, doi: [10.29297/orbit.v2i2.110](https://doi.org/10.29297/orbit.v2i2.110).
- McCord, S.E., Welty, J.L., Courtwright, J., Dillon, C., Traynor, A., Burnett, S.H., Courtwright, E.M., Fults, G., Karl, J.W., Webb, N.P., Tweedie, C. and Van Zee, J.W. (2022), "Ten practical questions to improve data quality", *Rangelands*, Vol. 44 No. 1, pp. 17-28, doi: [10.1016/j.rala.2021.07.006](https://doi.org/10.1016/j.rala.2021.07.006).
- Metcalfe, J. and Crawford, K. (2016), "Where are human subjects in big data research? The emerging ethics divide", *Big Data and Society*, Vol. 3 No. 1, 2053951716650211, doi: [10.1177/2053951716650211](https://doi.org/10.1177/2053951716650211).
- Mogaji, E., Soetan, T.O. and Kieu, T.A. (2020), "The implications of artificial intelligence on the digital marketing of financial services to vulnerable customers", *Australasian Marketing Journal*, Vol. 29 No. 3, pp. 235-242, doi: [10.1016/j.ausmj.2020.05.003](https://doi.org/10.1016/j.ausmj.2020.05.003).
- Mohammed, K., Ayesh, A. and Boiten, E. (2021), "Complementing privacy and utility trade-off with self-organising maps", *Cryptography*, Vol. 5 No. 3, p. 20, doi: [10.3390/cryptography5030020](https://doi.org/10.3390/cryptography5030020), available at: <https://www.mdpi.com/2410-387X/5/3/20>
- Morán-Reyes, A.A. (2022), "Towards an ethical framework about Big Data era: metaethical, normative ethical and hermeneutical approaches", *Heliyon*, Vol. 8 No. 2, e08926, doi: [10.1016/j.heliyon.2022.e08926](https://doi.org/10.1016/j.heliyon.2022.e08926).
- Muller, S.H.A., van Thiel, G., Vrana, M., Mostert, M. and van Delden, J.J.M. (2022), "Patients' and publics' preferences for data-intensive health research governance: survey study", *JMIR Hum Factors*, Vol. 9 No. 3, e36797, doi: [10.2196/36797](https://doi.org/10.2196/36797).
- Mullins, M., Holland, C.P. and Cunneen, M. (2021), "Creating ethics guidelines for artificial intelligence and big data analytics customers: the case of the consumer European insurance market", *Patterns*, Vol. 2 No. 10, 100362, doi: [10.1016/j.patter.2021.100362](https://doi.org/10.1016/j.patter.2021.100362).
- Munn, L. (2022), "The uselessness of AI ethics", *AI and Ethics*, Vol. 3 No. 3, pp. 1-9, doi: [10.1007/s43681-022-00209-w](https://doi.org/10.1007/s43681-022-00209-w).
- Nawawi, A. and Ahmad Saiful Azlin Puteh, S. (2018), "Whistle blowing intentions – evidence from Malaysian PLC", *International Journal of Law and Management*, Vol. 60 No. 5, pp. 1111-1125, doi: [10.1108/IJLMA-04-2017-0096](https://doi.org/10.1108/IJLMA-04-2017-0096).
- Nersessian, D. (2018), "The law and ethics of big data analytics: a new role for international human rights in the search for global standards", *Business Horizons*, Vol. 61 No. 6, pp. 845-854, doi: [10.1016/j.bushor.2018.07.006](https://doi.org/10.1016/j.bushor.2018.07.006).
- Nilashi, M., Baabdullah, A.M., Abumalloh, R.A., Ooi, K.-B., Tan, G.W.-H., Giannakis, M. and Dwivedi, Y.K. (2023), "How can big data and predictive analytics impact the performance and competitive advantage of the food waste and recycling industry?", *Annals of Operations Research*, Vol. 348 No. 3, pp. 1649-1690, doi: [10.1007/s10479-023-05272-y](https://doi.org/10.1007/s10479-023-05272-y).
- Olesen, T. (2019), "The politics of whistleblowing in digitalized societies", *Politics and Society*, Vol. 47 No. 2, pp. 277-297, doi: [10.1177/0032329219844140](https://doi.org/10.1177/0032329219844140).

- Olszak, C. and Zurada, J. (2019), "Big data-driven value creation for organizations", *Paper presented at the Proceedings of the 52nd Hawaii International Conference on System Sciences*.
- Omri, N., Al Masry, Z., Mairot, N., Giampiccolo, S. and Zerhouni, N. (2021), "Towards an adapted PHM approach: data quality requirements methodology for fault detection applications", *Computers in Industry*, Vol. 127, 103414, doi: [10.1016/j.compind.2021.103414](https://doi.org/10.1016/j.compind.2021.103414).
- Ongsakul, V. and Sen, S.K. (2018), "Creating shared value through big-data analytics: a conceptual treatise on sustainability presence", *Journal of Business and Retail Management Research*, Vol. 12 No. 4, doi: [10.24052/jbrmr/v12is04/art-10](https://doi.org/10.24052/jbrmr/v12is04/art-10).
- Paul, J. and Feliciano-Cestero, M.M. (2021), "Five decades of research on foreign direct investment by MNEs: an overview and research agenda", *Journal of Business Research*, Vol. 124, pp. 800-812, doi: [10.1016/j.jbusres.2020.04.017](https://doi.org/10.1016/j.jbusres.2020.04.017).
- Polding, R.D. and Dieguez, M.E. (2021), "An investigation into the effectiveness of big data in organizations, the use of customer data, and the ethical implications of the data economy", *Proceedings of the 2021 International Symposium on Electrical, Electronics and Information Engineering (ISEEIE 2021)*, Seoul, Republic of Korea, Association for Computing Machinery, doi: [10.1145/3459104.3459201](https://doi.org/10.1145/3459104.3459201).
- Power, D., Heavin, C. and O'Connor, Y. (2021), "Balancing privacy rights and surveillance analytics: a decision process guide", *Journal of Business Analytics*, Vol. 4, pp. 1-16, doi: [10.1080/2573234X.2021.1920856](https://doi.org/10.1080/2573234X.2021.1920856).
- Provost, F. and Fawcett, T. (2013), "Data science and its relationship to big data and data-driven decision making", *Big Data*, Vol. 1 No. 1, pp. 51-59, doi: [10.1089/big.2013.1508](https://doi.org/10.1089/big.2013.1508).
- Randolph-Seng, B., Atinc, Y., Bui, S. and Malik, Z. (2024), "Big data and the disillusioned employee: inside the ethical internet of things", *Technological Forecasting and Social Change*, Vol. 201, 123271, doi: [10.1016/j.techfore.2024.123271](https://doi.org/10.1016/j.techfore.2024.123271).
- Rautdesai, R., Nandekar, U., Kedari, A. and Patil, Y. (2019), "Big data and privacy-a legal perspective and comparative study of the USA and India", *International Journal of Process Management and Benchmarking*, Vol. 9 No. 2, pp. 250-275, doi: [10.1504/ijpmb.2019.099334](https://doi.org/10.1504/ijpmb.2019.099334).
- Recknagel, F. and Michener, W.K. (2017), *Ecological Informatics: Data Management and Knowledge Discovery*, 3rd ed., Springer, Cham.
- Roche, J. and Jamal, A. (2021), "A systematic literature review of the role of ethics in big data", *Paper Presented at the Cybersecurity, Privacy and Freedom Protection in the Connected World: Proceedings of the 13th International Conference on Global Security, Safety and Sustainability*, London, January 2021, pp. 327-342, doi: [10.1007/978-3-030-68534-8_20](https://doi.org/10.1007/978-3-030-68534-8_20).
- Rodgers, W., Murray, J.M., Stefanidis, A., Degbey, W.Y. and Tarba, S.Y. (2023), "An artificial intelligence algorithmic approach to ethical decision-making in human resource management processes", *Human Resource Management Review*, Vol. 33 No. 1, 100925, doi: [10.1016/j.hrmr.2022.100925](https://doi.org/10.1016/j.hrmr.2022.100925).
- Rubeis, G. (2022), "iHealth: the ethics of artificial intelligence and big data in mental healthcare", *Internet Interventions*, Vol. 28, 100518, doi: [10.1016/j.invent.2022.100518](https://doi.org/10.1016/j.invent.2022.100518).
- Ryan, M. (2020), "Agricultural big data analytics and the ethics of power", *Journal of Agricultural and Environmental Ethics*, Vol. 33 No. 1, pp. 49-69, doi: [10.1007/s10806-019-09812-0](https://doi.org/10.1007/s10806-019-09812-0).
- Ryan, M., Antoniou, J., Brooks, L., Jiya, T., Macnish, K. and Stahl, B. (2020), "The ethical balance of using smart information systems for promoting the united nations' sustainable development goals", *Sustainability*, Vol. 12 No. 12, p. 4826, doi: [10.3390/su12124826](https://doi.org/10.3390/su12124826), available at: <https://www.mdpi.com/2071-1050/12/12/4826>
- Ryen, A. (2023), "Ethics and politics in qualitative research", in Tierney, R.J., Rizvi, F. and Ercikan, K. (Eds), *International Encyclopedia of Education*, 4th ed., Elsevier, Oxford, pp. 33-44.
- Salerno, J., Knoppers, B.M., Lee, L.M., Hlaing, W.M. and Goodman, K.W. (2017), "Ethics, big data and computing in epidemiology and public health", *Annals of Epidemiology*, Vol. 27 No. 5, pp. 297-301, doi: [10.1016/j.annepidem.2017.05.002](https://doi.org/10.1016/j.annepidem.2017.05.002).

- Santoro, G., Fiano, F., Bertoldi, B. and Ciampi, F. (2018), "Big data for business management in the retail industry", *Management Decision*, Vol. 57 No. 8, pp. 1980-1992, doi: [10.1108/md-07-2018-0829](https://doi.org/10.1108/md-07-2018-0829).
- Schwartz, M.S. (2016), "Ethical decision-making theory: an integrated approach", *Journal of Business Ethics*, Vol. 139 No. 4, pp. 755-776, doi: [10.1007/s10551-015-2886-8](https://doi.org/10.1007/s10551-015-2886-8).
- Seele, P. (2017), "Predictive sustainability control: a review assessing the potential to transfer big data driven 'predictive policing' to corporate sustainability management", *Journal of Cleaner Production*, Vol. 153, pp. 673-686, doi: [10.1016/j.jclepro.2016.10.175](https://doi.org/10.1016/j.jclepro.2016.10.175).
- Shafiq, A., Ahmed, M.U. and Mahmoodi, F. (2020), "Impact of supply chain analytics and customer pressure for ethical conduct on socially responsible practices and performance: an exploratory study", *International Journal of Production Economics*, Vol. 225, 107571, doi: [10.1016/j.ijpe.2019.107571](https://doi.org/10.1016/j.ijpe.2019.107571).
- Speckmann, F. (2021), "Web scraping: a useful tool to broaden and extend psychological research", *Zeitschrift für Psychologie*, Vol. 229 No. 4, pp. 241-244, doi: [10.1027/2151-2604/a000470](https://doi.org/10.1027/2151-2604/a000470).
- Stefanizzi, S. (2022), "From a theory-centric view of social research to a data-centric approach", *Italian Sociological Review*, Vol. 12 No. 7S, pp. 651-663, available at: <http://search.proquest.com/qulib.idm.oclc.org/scholarly-journals/theory-centric-view-social-research-data-approach/docview/2703522054/se-2?accountid=13370>
- Stylianou, A. and Talias, M. (2016), "Big data in healthcare: a discussion on the big challenges", *Health Technology*, Vol. 7 No. 1, pp. 97-107, doi: [10.1007/s12553-016-0152-4](https://doi.org/10.1007/s12553-016-0152-4).
- Tanggamani, A.R. and Alias, N.A. (n.d.), "Big data: balancing its usefulness in managing an organization performance and ethical consideration".
- Tien, J.M. (2020), "Toward the fourth industrial revolution on real-time customization", *Journal of Systems Science and Systems Engineering*, Vol. 29 No. 2, pp. 127-142, doi: [10.1007/s11518-019-5433-9](https://doi.org/10.1007/s11518-019-5433-9).
- Ulltveit-Moe, N. (2014), "A roadmap towards improving managed security services from a privacy perspective", *Ethics and Information Technology*, Vol. 16 No. 3, pp. 227-240, doi: [10.1007/s10676-014-9348-3](https://doi.org/10.1007/s10676-014-9348-3).
- van Maanen, G. (2022), "AI ethics, ethics washing, and the need to politicize data ethics", *Digital Society*, Vol. 1 No. 2, p. 9, doi: [10.1007/s44206-022-00013-3](https://doi.org/10.1007/s44206-022-00013-3).
- Van Rooijen, M. (2015), *Metaethics: A Contemporary Introduction*, Routledge.
- Vayena, E. and Tasioulas, J. (2016), "The dynamics of big data and human rights: the case of scientific research", *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, Vol. 374 No. 2083, 20160129, doi: [10.1098/rsta.2016.0129](https://doi.org/10.1098/rsta.2016.0129).
- Vial, G. (2023), "Data governance and digital innovation: a translational account of practitioner issues for IS research", *Information and Organization*, Vol. 33 No. 1, 100450, doi: [10.1016/j.infoandorg.2023.100450](https://doi.org/10.1016/j.infoandorg.2023.100450).
- Wamba, S.F., Gunasekaran, A., Akter, S., Ren, S.J.-f., Dubey, R. and Childe, S.J. (2017), "Big data analytics and firm performance: effects of dynamic capabilities", *Journal of Business Research*, Vol. 70, pp. 356-365, doi: [10.1016/j.jbusres.2016.08.009](https://doi.org/10.1016/j.jbusres.2016.08.009).
- Wang, M., Li, Y., Pan, T. and Jia, N. (2024), "Plant natural compounds in the cancer treatment: a systematic bibliometric analysis", *Heliyon*, Vol. 10 No. 14, e34462, doi: [10.1016/j.heliyon.2024.e34462](https://doi.org/10.1016/j.heliyon.2024.e34462).
- Webb, H.W. and Webb, L.A. (2004), "SiteQual: an integrated measure of Web site quality", *Journal of Enterprise Information Management*, Vol. 17 No. 6, pp. 430-440, doi: [10.1108/17410390410566724](https://doi.org/10.1108/17410390410566724).
- Webley, S. (2001), "Business ethics: a SWOT exercise", *Business Ethics: A European Review*, Vol. 10 No. 3, pp. 267-271, doi: [10.1111/1467-8608.00240](https://doi.org/10.1111/1467-8608.00240).
- Winickoff, D.E., Jamal, L. and Anderson, N.R. (2016), "New modes of engagement for big data research", *Journal of Responsible Innovation*, Vol. 3 No. 2, pp. 169-177, doi: [10.1080/23299460.2016.1190443](https://doi.org/10.1080/23299460.2016.1190443).

- Xafis, V., Schaefer, G.O., Labude, M.K., Brassington, I., Ballantyne, A., Lim, H.Y., Lipworth, W., Lysaght, T., Stewart, C., Laurie, G.T., Tai, E.S. and Sun, S. (2019), "An ethics framework for big data in health and research", *Asian Bioethics Review*, Vol. 11 No. 3, pp. 227-254, doi: [10.1007/s41649-019-00099-x](https://doi.org/10.1007/s41649-019-00099-x).
- Yadav, N., Kumar, R. and Malik, A. (2022), "Global developments in coopetition research: a bibliometric analysis of research articles published between 2010 and 2020", *Journal of Business Research*, Vol. 145, pp. 495-508, doi: [10.1016/j.jbusres.2022.03.005](https://doi.org/10.1016/j.jbusres.2022.03.005).
- Yalcinkaya, T. and Cinar Yücel, S. (2024), "Bibliometric and content analysis of ChatGPT research in nursing education: the rabbit hole in nursing education", *Nurse Education in Practice*, Vol. 77, 103956, doi: [10.1016/j.nepr.2024.103956](https://doi.org/10.1016/j.nepr.2024.103956).
- Yallop, A. and Seraphin, H. (2020), "Big data and analytics in tourism and hospitality: opportunities and risks", *Journal of Tourism Futures*, Vol. 6 No. 3, pp. 257-262, doi: [10.1108/JTF-10-2019-0108](https://doi.org/10.1108/JTF-10-2019-0108).
- Yallop, A.C., Gică, O.A., Moisescu, O.I., Coroş, M.M. and Séraphin, H. (2023), "The digital traveller: implications for data ethics and data governance in tourism and hospitality", *Journal of Consumer Marketing*, Vol. 40 No. 2, pp. 155-170, doi: [10.1108/JCM-12-2020-4278](https://doi.org/10.1108/JCM-12-2020-4278).
- Yazdani, N. and Murad, H.S. (2015), "Toward an ethical theory of organizing", *Journal of Business Ethics*, Vol. 127 No. 2, pp. 399-417, doi: [10.1007/s10551-014-2049-3](https://doi.org/10.1007/s10551-014-2049-3).
- Ye, C. (2024), "The current state of big data analytics research in marketing: a systematic review using TCCM approach", *Journal of Global Scholars of Marketing Science*, Vol. 34 No. 3, pp. 393-415, doi: [10.1080/21639159.2023.2275792](https://doi.org/10.1080/21639159.2023.2275792).
- Zhu, C. (2022), "Construction and risk analysis of marketing system based on AI", *Scientific Programming*, Vol. 2022, pp. 1-9, doi: [10.1155/2022/2839834](https://doi.org/10.1155/2022/2839834).
- Zhu, L., Ou, L., Yang, Y., Zhao, D., Liu, B., Liu, R. and Feng, H. (2024), "A bibliometric and visualised analysis of proliferative verrucous leucoplakia from 2003 to 2023", *International Dental Journal*, Vol. 75 No. 1, pp. 333-344, doi: [10.1016/j.identj.2024.07.001](https://doi.org/10.1016/j.identj.2024.07.001).
- Zimmer, M. (2018), "Addressing conceptual gaps in big data research ethics: an application of contextual integrity", *Social Media+ Society*, Vol. 4 No. 2, doi: [10.1177/2056305118768300](https://doi.org/10.1177/2056305118768300).
- Zulkarnain, N., Anshari, M., Hamdan, M. and Fithriyah, M. (2021), "Big data in business and ethical challenges", *Paper Presented at the 2021 International Conference on Information Management and Technology (ICIMTech)*, pp. 298-303, doi: [10.1109/icimtech53080.2021.9534963](https://doi.org/10.1109/icimtech53080.2021.9534963).

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